Ownership Rights as Innovation Policy: Evidence on the Direction of Innovation

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Abstract

This paper studies how the allocation of intellectual-property rights over publicly funded inventions shapes the direction of private innovation and the diffusion of technological knowledge. I exploit the 1983 Presidential Memorandum on Government Patent Policy, which expanded contractors' ability to retain ownership of future federally funded inventions while leaving the underlying research funding mechanism largely unchanged. Using a matched difference-in-differences design at the firm–technology level, I find that stronger contractor ownership reduced cross-technology spillovers by approximately 25–30 percent, lowered the share of technologically similar follow-on innovation occurring outside the originating firm, and increased private patent market value by 30–40 percent. The evidence indicates that stronger ownership rights changed the composition rather than the quantity of innovation, shifting research toward inventions with greater private appropriability but narrower external applicability. The findings identify ownership allocation as an important innovation-policy instrument, showing that the allocation of rights over publicly funded research shapes not only firms' incentives to innovate but also the social reach of the innovations they choose to pursue.

1 Introduction

Public R&D policy faces a fundamental trade-off. Granting firms ownership of publicly funded inventions may strengthen incentives for commercialization by increasing private appropriability, but it may also reduce the diffusion of knowledge generated by publicly financed research. Despite decades of debate surrounding Bayh-Dole and related reforms, we know surprisingly little about how ownership rights affect the type of innovation firms produce. This paper combines newly assembled ownership data with a previously unexplored federal patent-policy reform to provide causal evidence on that question.

I study the 1983 Presidential Memorandum on Government Patent Policy, which sharply expanded contractors' rights to retain title over federally funded inventions. Exploiting this policy change together with newly assembled data on patent ownership and firm-level innovation, I estimate how stronger contractor ownership affected the social and private returns to federally funded research. The findings document a clear pattern; contractor ownership substantially reduces the diffusion of knowledge while increasing the private value of innovation.

Despite the large literature on spillovers, innovation, and growth, the role of patent ownership remains unresolved for at least two reasons. First, there is limited and often inconsistent data on both whether a patent resulted from federal support and who held title to that invention. Although patents are required to disclose government support, noncompliance remains substantial, and the absence of a meaningful monitoring mechanism has led to persistent under-reporting[1]. Even among patents that do disclose federal funding, existing sources often report conflicting ownership status: some list the contractor, others the agency, some both, and some neither. Second, identifying the causal effect of patent rights on innovation outcomes is inherently difficult because it requires a credible counterfactual for the path of innovation under an alternative ownership regime.

This paper addresses both challenges by combining newly assembled data on the ownership status of federally funded patents with a policy change that sharply expanded contractor title rights in 1983. I use administrative-data-based ownership records from Gross and Sampat (2025), which resolve the inconsistencies in earlier ownership sources, and combine them with the patent-to-firm links in Dyevre (2023), which, to my knowledge, are the most comprehensive publicly available source for linking patents to firms at the scale required for this analysis. Exploiting the 1983 presidential memorandum as a source of plausibly exogenous variation, I study how contractor

[1] see, for example, in Rai and Sampat (2012)

ownership affects the direction and quality of innovation.

Identifying the effect of contractor patent ownership is challenging for two reasons. First, firms receiving federal R&D support differ systematically from firms that do not, including in scale, research intensity, and innovative capacity. Second, under the pre-reform regime, agencies could retain title selectively, so ownership of federally funded inventions was not randomly assigned. I address the first concern by constructing a matched control group within narrow technology-industry cells based on pre-policy characteristics and outcome trends. I then estimate a difference-in-differences specification at the firm-technology level with firm-technology fixed effects, technology-year fixed effects, and industry-year fixed effects, while allowing baseline observables to follow flexible time paths. The 1983 presidential memorandum is useful in this setting because it generated a rapid, cross-agency shift toward contractor title, reducing the role of agency-specific patent policies.

A remaining concern is that agencies may have retained title before 1983 for inventions with systematically different characteristics. However, the relevant margin for this study excludes inventions subject to persistent national-security or similar exemptions, which were less likely to be affected by the reform. Among the inventions whose ownership status plausibly changed, pre-reform agency retention may, if anything, have been concentrated among patents with broader applicability or weaker private appropriability, that is, patents with relatively high ex ante spillover potential. Under that selection pattern, the post-reform decline in spillovers is difficult to attribute to selective retention and, if anything, suggests that the estimated effect is conservative.

The empirical findings point to a clear tradeoff between private appropriation and knowledge diffusion. Following the reform, cross-technology knowledge spillovers from treated firm-technology units decline by about 25 to 30 percent relative to comparable untreated units, indicating a substantial decline in downstream technological diffusion. Text-based measures of patent significance proposed by Kelly et al. (2021) also fall by a similar magnitude, while a text-based externality measure capturing the share of forward technological similarity accruing outside the originating firm declines by about 16 to 18 percent. In contrast, the stock-market-based measure of patented inventions of Kogan et al. (2017) increases by roughly 40 percent. Critically, the institutional setting helps distinguish changes in knowledge diffusion from changes in the nature of innovation itself. The 1983 reform expanded contractors' ownership rights but did not alter the disclosure requirements of the patent system: patented inventions continued to become publicly available upon grant. Thus, the mechanism through which patented knowledge became publicly accessible remained unchanged. Taken together, the evidence indicate that contractor ownership does not simply change who captures the returns to innovation; it changes the type of innovation that is produced.

The average treatment effect masks substantial heterogeneity across firms. While the ownership change results in narrower portfolio of inventions for both high and low R&D spending firms, it does so in completely different ways. Among firms with relatively limited pre-treatment R&D resources, contractor ownership reduces the production of inventions that generate the largest knowledge spillovers. By contrast, firms with substantial research capabilities expand patenting in the lower half of the spillover distribution, while exhibiting no corresponding increase in high-spillover inventions. The evidence is consistent with two complementary mechanisms: a deterioration in the production of high-spillover inventions among resource-constrained firms and a dilution of innovation among large R&D performers through increased production of more privately appropriable research. The stock-market gains are concentrated entirely among firms in the highest quartile of the R&D expenditure distribution, suggesting substantially larger private-return gains than implied by the average treatment effect.

The takeaway is that across citation-based, text-based, and market-based measures, the evidence points to the same conclusion: stronger ownership rights redirect innovation toward outcomes with greater private appropriability but weaker external knowledge diffusion. The conclusion is robust across alternative measures of innovation quality and alternative approaches to constructing the counterfactual, including the synthetic difference-in-differences estimator of Arkhangelsky et al. (2021).

This paper makes three contributions: First, it provides causal evidence that the ownership status of publicly funded invention shapes not only the commercialization of research, but also the direction of research itself. Second, it contributes to the literature on innovation policy and the assignment of intellectual property rights by identifying a tradeoff that has received less attention: stronger contractor control may narrow the private-social return wedge not only by increasing private appropriation, but also by reducing an important component of the social return to publicly-funded R&D, namely the diffusion of knowledge spillovers. Third, the results show that ownership rights reallocate innovative effort differently depending on firm's absorptive capacity: firms with stronger R&D capabilities are better able to convert ownership rights into privately valuable innovation, whereas firms with weaker capabilities reduce activity precisely in the segment of research that generates the largest external benefits.

Taken together, the results highlight a central tension in innovation policy. Knowledge spillovers constitute one of the principal channels through which innovation generates social returns and, consequently, one of the central justifications for public support of R&D in modern theories of innovation and endogenous growth. Institutions designed to strengthen incentives for commercialization may therefore weaken an important mechanism through which R&D contributes to long-run growth in two ways. First, by redirecting research away from inventions with the largest technological spillovers, they reduce both the social returns to innovation and the opportunities for other firms to benefit from externally generated knowledge. Second, by shifting innovative effort toward more narrowly applicable research, they may gradually narrow firms' effective knowledge base, limiting their ability to recognize, combine, and exploit external technological advances in the future.

Related Literature. This paper relates most directly to the literature studying the private and social returns to innovation. Beginning with Arrow (1962) and extending through Bloom et al. (2013) and Jones and Summers (2022), this literature emphasizes that spillovers from R&D create a wedge between private and social returns, and while proposing different methods for calculating spillovers they all agree that social returns exceed private by a wide margin[1]. This branch of the literature focuses on corporate-R&D and the private return documenting both positive spillovers, through technology, and negative from product-market rivalry. The current paper extends that literature by identifying an additional institutional margin operating through publicly funded R&D: the allocation of ownership rights over the resulting inventions. I find that contractor ownership of a patent's rights not only increases their private return but also reduces its social return by changing the direction of innovative effort.

The second branch that this paper is especially close to is that estimating the social returns to publicly funded R&D. A long empirical literature focusing on agency-specific or case-studies shows that government-funded research generates substantial downstream innovation and productivity gains[2]. Recent contributions using cross-agency or aggregate government variation include Field-

[1] for example, Lucking et al. (2019) find social-to-private return ratio to be 4-to-1 while Jones and Summers (2022) raise it to 5-to-1 under their conservative scenario

[2] Azoulay et al. (2019) look on NIH funding, Myers and Lanahan (2022) on DOE, Moretti et al. (2025) or Howell et al. (2025) on DoD. Azoulay et al. (2019) find a $10 million increase in funding to generate three to four more patents; two of which on different technology areas, Myers and Lanahan (2022) confirm this finding reporting three additional patents being generated for each patent funded, Moretti et al. (2025) report a 5% to 6% increase in privately funded R&D for each 10% increase in government-financed R&D, Howell et al. (2025) find that winning a funding award from US Air Force through the Small Business Innovation Research (SBIR) increases the probability of subsequent venture-capital investment by 128%, having a patent by 79% and a high-originality patent by 194%

house and Mertens (2023), Dyevre (2023), or Fieldhouse and Mertens (2026) and they re-affirm the strong gains. My paper complements this work in two ways: first, by introducing a new source of identifying variation: a federal policy change that altered the ownership of publicly funded inventions across agencies. Second, by showing that public R&D policy has two policy margins: the scale of public funding and the ownership of the resulting knowledge. The literature has devoted considerable attention to the former; I study the latter.

A third related part of the literature is that discussing the changing direction of private-sector research. Budish et al. (2015) show that private research can be distorted away from long-horizon projects, while Arora et al. (2021) document a shift in corporate innovation away from research and toward development. My findings are complementary to that evidence. The analysis shows that when ownership rights over public-funded inventions are strengthened, firms redirect innovative activity toward outcomes with greater private appropriability but weaker knowledge diffusion. The paper therefore provides causal evidence on one institutional channel through which the balance between research-oriented and development-oriented innovation can shift.

Finally, the paper relates to the literature on patents as an intellectual property strategy and research investments. As emphasized by Williams (2017), credible causal evidence on how patent rights affect innovation remains limited, largely because it is difficult to observe a convincing counterfactual path of technological development[1]. My paper addresses that challenge by exploiting a sharp change in federal patent policy and combining it with a matched difference-in-differences design. This provides, novel causal evidence that strengthening firms' ownership rights over publicly funded inventions affects not only the private value of innovation, but also its social value through knowledge spillovers. More broadly, the results suggest that patent policy should be evaluated not only in terms of whether it strengthens incentives to innovate, but also in terms of how it changes the composition of innovation between privately appropriable and socially diffusive research.

[1] An important exemption is that of Lerner (2009) who attempted to identify the effect of stronger patent property rights on a country scale and found no effect on R&D spending. A separate sub-branch studies the effect of patent litigation on future innovation (eg Galasso and Schankerman (2018)). From the management literature, recent work of Masclans et al. (2025) builds a measure of "commercial potential" of science using text-processing algoriths on abstracts of scientific articles. For comparative studies on intellectual property strategies of firms, see for example Mezzanotti and Simcoe (2023)

2 Conceptual Framework

Technological knowledge has long been understood as a public good because it is nonrival and only imperfectly excludable. The benefits of that were formally expressed in Romer (1986), who argued that knowledge spillovers from investment can generate increasing returns at the aggregate level, helping launch the new growth literature. Modern endogenous growth theory still places technological progress at the center of long-run growth, with own-R&D and knowledge spillovers as its main engine (Romer, 1990; Akcigit et al., 2021).

The semi-public nature of knowledge creates a fundamental tension for firms. Because they cannot fully appropriate the returns to innovation, private incentives to invest fall short of the social optimum, particularly for research whose benefits diffuse widely (Nelson, 1959). This market failure motivates policies aimed at narrowing the private-social return wedge, including R&D subsidies, tax incentives, and direct public support for research.

From the perspective of social welfare, imperfect appropriability implies that the social return to innovation exceeds the private return, creating the well-known private-social wedge (Arrow, 1962). From the firm's perspective, however, outgoing spillovers is a negative externality benefiting rivals and reducing the profitability of innovation through business stealing (Aghion, 1992; Bloom et al., 2013; Lucking et al., 2019). Although these approaches differ in their interpretation of spillovers, both ultimately originate from the same underlying tension between knowledge as a semi-public good and firms' incentives to privately appropriate its returns.

A similar tension appears in theories of firms' own research investment. Incoming spillovers complement own-R&D[1] and own-R&D strengthens firms' absorptive capacity of spillovers (Cohen et al., 1990), while outgoing spillovers weaken incentives to invest by facilitating competitors' innovation (Arora et al., 2021). Again, the contrasting role assigned to spillovers reflects the same underlying conflict between socially valuable knowledge diffusion and private appropriation.

The common feature of these theories is that they characterize the consequences of imperfect appropriability once the institutional environment is taken as given. As ownership in corporate-R&D is typically already determined before research projects are undertaken, it affects the expected private return to alternative research trajectories before innovation occurs. Ownership therefore influences not only the incentive to invest in R&D, but also the direction of innovative effort. This paper focuses on one institution that directly governs appropriability itself: the allocation of

[1] several studies document strong complementarities with own-R&D. see, for example, Aghion and Jaravel (2015)

ownership rights over publicly funded inventions.

Publicly funded patents provide a natural setting for studying this institutional margin because a change in the degree of appropriation is clearly identifiable while the source of funding is held fixed. The 1983 Presidential Memorandum is particularly well suited for identifying this mechanism because ownership rights were reassigned prospectively. Firms therefore knew, before undertaking research, whether successful inventions would remain under their control, allowing ownership expectations to influence project selection, commercialization incentives, and ultimately the type of innovation produced.

The empirical analysis yields three implications.

Implication 1. (The Trade-off)

The estimates reveal that increasing appropriability is accompanied by lower knowledge diffusion. Rather than increasing one component of the return to innovation, stronger ownership rights induce a trade-off between private appropriation and the diffusion of socially valuable knowledge.

Implication 2. (Ownership as a Policy Instrument)

The findings imply that the private-social wedge is itself endogenous to institutional design. Traditionally, innovation policy seeks to narrow this wedge by increasing the amount of research undertaken through subsidies and public funding. The evidence here highlights an additional policy margin: the allocation of ownership rights over publicly funded inventions. A narrower private-social wedge need not arise solely from an increase in private returns; it may also reflect a decline in the social return generated by the resulting innovation. These two mechanisms have fundamentally different welfare implications. Ownership should therefore be viewed as a policy instrument determining where publicly funded innovation is located along the trade-off between private appropriability and knowledge diffusion.

Implication 3. (Dynamic Implications)

A further implication follows from absorptive-capacity theory. If ownership affects the direction of innovative activity, it also shapes the knowledge base accumulated within firms and, consequently, their future ability to recognize and exploit external knowledge. The effects of ownership rights may therefore extend beyond the contemporaneous allocation of innovative effort to influence the dynamic evolution of firms' innovative capabilities.

3 Historical & Institutional Background

The U.S. debate over patent rights in publicly funded research emerged from the broader postwar challenge of reconversion from war to peace. As World War II ended, policymakers faced two closely related questions: how to convert the scientific and industrial capacity built during the war into peacetime prosperity, and what role the federal government should play in supporting innovation. Wartime experience had demonstrated the value of large-scale public investment in research, strengthening the case for a continuing federal role. At the same time, the emerging Cold War strengthened opposition to centralized federal control over science and a more egalitarian distribution of public research funding as such policies were increasingly associated with state planning rather than the market-oriented model policymakers sought to preserve. The central debate was therefore not whether the government should support research, but rather how that support should be organized and who should benefit from the resulting discoveries.

This tension is reflected in the *Bush–Kilgore debate*.[1] Bush emphasized scientific autonomy and competitive allocation of research funds, whereas Kilgore favored stronger public oversight and was more concerned about the private appropriation of publicly financed discoveries.[2] Yet both positions reflected the same historical reality: wartime experience had made a permanent federal role in research difficult to deny.

The compromise that emerged represented a fundamental departure from prewar innovation policy. Before World War II, civilian research was conducted largely in private industrial laboratories and universities supported by industry and philanthropic foundations, while federal involvement remained limited out of concern that government funding might restrict scientific freedom. Postwar policymakers abandoned this reluctance toward federal support, but they did not centralize the research system itself. Instead, research administration remained decentralized, with individual agencies retaining substantial discretion over funding, contracting, and patent policy. This agency-by-agency approach would shape U.S. innovation policy for the following three decades and ultimately become central to the evolution of federal patent rights.

[1] Vannevar Bush, director of the Office of Scientific Research and Development (OSRD), argued in *Science, the Endless Frontier* (1945) for a merit-based research system insulated from day-to-day political influence. Senator Harley Kilgore instead favored stronger public oversight and expressed concerns about the concentration of publicly funded research within a small number of universities and large corporations. Today, Bush's report is widely regarded as the blueprint for the postwar U.S. innovation system.

[2] The NSF's official history identifies patent ownership, administrative control, and the balance between basic and applied research as the principal points of contention in these debates.

Against that backdrop, a central policy question emerged: who should own inventions created with federal support? In the immediate postwar years, the prevailing answer favored government ownership, reflecting concerns that granting exclusive rights to individual contractors could amount to a public subsidy benefiting selected firms[1]. Over time, however, dissatisfaction grew with the fragmented agency-by-agency patent regime. By the late 1970s, policymakers increasingly argued that federally funded inventions were not being developed or commercialized efficiently under a system in which agencies frequently retained title and licensed inventions nonexclusively. A 1998 Government Accountability Office review summarizes this contemporary diagnosis: before Bayh–Dole there was "no uniform federal policy" governing patents arising from federally funded research, and by 1980 fewer than five percent of roughly 28,000 federally owned patents had been licensed, compared with substantially higher licensing rates where contractors retained title.

The movement toward contractor ownership was gradual rather than abrupt. President Kennedy's 1963 Statement of Government Patent Policy established the first government-wide framework governing ownership of federally funded inventions. While government ownership remained the general presumption, agencies were allowed to grant contractors greater rights where doing so was expected to facilitate practical application. President Nixon's 1971 revision extended this commercialization-oriented approach by giving agencies broader discretion to permit contractors stronger ownership rights whenever this was expected to encourage utilization[2].

Bayh–Dole marked the next major step in this evolution. Enacted in 1980, the Act reversed the presumption of government ownership for federally funded inventions made by universities, nonprofit organizations, and small businesses. Its statement of policy is revealing. Congress presented patent rights as policy tools for promoting commercialization and collaboration with industry, but also higher employment and productivity growth[3]. The first key innovation of Bayh–Dole was

[1] The *National Patent Planning Commission* (1945) advocated government ownership of publicly funded inventions while allowing broad commercial licensing. Similarly, the Attorney General's 1947 Report to the President argued for a uniform federal policy and government ownership of federally funded inventions in order to *"avoid government favoritism toward particular firms"* (Eisenberg, 1996)

[2] As Nixon's memorandum stated, the objective was to provide agencies with *"additional authority to permit contractors to obtain greater rights to inventions where necessary to achieve utilization,"* while improving the government's ability to evaluate the effectiveness of federal patent policy (August 23, 1971, the Nixon Memorandum on Government Patent Policy)

[3] Specifically, 35 U.S.C. §200 states that the patent system should be used "to promote the utilization" of federally supported inventions and "to promote the commercialization and public availability" of inventions made in the United States, while preserving sufficient government rights to protect the public against nonuse or unreasonable use

that it established the first government-wide statutory framework governing ownership of federally funded inventions. Rather than relying on agency-specific patent policies, the Act created a common set of ownership rules for all research performers within its scope, drastically reducing institutional fragmentation.

Equally important, Bayh–Dole changed the timing of ownership decisions for the first time. Rather than being negotiated after an invention had been produced, ownership became part of the research contract itself. Contractors disclose subject inventions to the funding agency, elect whether to retain title within statutory deadlines, and pursue patent protection under a standardized framework. From an economic perspective, this prospective allocation of ownership matters because it shapes incentives before research is undertaken. By reducing uncertainty over the appropriation of future returns, the ownership regime can influence research planning, project selection, organizational investment, and ultimately the type of inventions firms choose to pursue. Patent policy therefore affects not only the commercialization of existing discoveries but also the direction of innovative activity itself.

Bayh–Dole, however, did not establish a uniform ownership regime across all federally funded research. Its provisions applied exclusively to universities, nonprofit organizations, and small businesses, leaving the large industrial contractors under the earlier agency-specific framework. That distinction changed with President Reagan's February 18, 1983 Memorandum on Government Patent Policy, which extended Bayh–Dole's contractor-oriented philosophy to all federally funded performers[1]. Although the movement toward contractor ownership had unfolded gradually over several decades, the 1983 memorandum represents a distinct institutional break by extending a common ownership regime to nearly all of the federal research system.

Figure 1 places this reform in historical perspective. Panel (a) shows the evolution of federally funded patents, while Panel (b) reports the share of those patents that title rights were held by the contractor (*"licensed"*)[2]. The figures illustrate the gradual shift toward contractor ownership reflected in successive postwar reforms, while also highlighting the discrete institutional change represented by the 1983 memorandum. It is this expansion of contractor ownership beyond the groups originally covered by Bayh–Dole that provides the basis for the empirical strategy developed in this paper.

[1]The 1984 Trademark Clarification Act subsequently codified this change by replacing the statutory reference to the 1971 Nixon policy

[2]To separate from patents that Agencies held title rights I name those whose rights are held by contractors as *license* patents and those held by agencies as *title* patents following Gross and Sampat (2025)

(a) Public-funded patents

(b) Licensed patents

Figure 1: Patent Funding & Licensing Timeline

Taken together, postwar U.S. innovation policy became increasingly guided by a commercialization mandate and a gradual strengthening of contractor property rights. The underlying premise was that allowing firms to appropriate a larger share of the returns to federally funded inventions would encourage commercialization and crowding-in of private investment in research thereby translating public research into productivity growth. Yet, stronger property rights may also reduce the diffusion of knowledge by redirecting effort away from projects with the largest external benefits, weakening the spillovers that constitute one of the principal engines of productivity growth. Whether expanding contractor ownership successfully promoted commercialization without reducing knowledge diffusion remained largely an empirical question. This paper examines that question.

4 Data

This paper combines several complementary datasets to study how changes in the ownership of federally funded inventions affect firms' innovation strategies and the social value of their research. The analysis links detailed information on patent ownership, government funding, citations, textual characteristics, and firm financials to construct a panel of firm–technology observations spanning the evolution of U.S. corporate innovation from 1950 to 2019. The empirical analysis focuses on the 1976–1995 period, which provides a sufficiently long pre- and post-policy window around the 1983 reform while avoiding later institutional changes in the patent system that could confound identification. The 1950–2019 sample is retained for descriptive evidence to document the broader trend. Appendix A provides additional details for both data sources and sample construction.

4.1 Data sources

The empirical analysis draws on several complementary datasets. The starting point is the patent-level ownership and government funding data assembled by Gross and Sampat (2025), which provide the most comprehensive publicly available record of federally funded U.S. patents. Relative to earlier sources, their data substantially improve ownership classification by exploiting newly uncovered administrative records from the USPTO archives that identify whether title to each federally funded invention remained with the government or was assigned to the contractor, and, for a subset of patents, identify the corresponding funding agency.

I combine these ownership records with the patent-to-firm linkage developed by Dyevre (2023), which matches U.S. patents to publicly listed Compustat firms. Because identifying the firms affected by the policy change is central to the empirical analysis, I extend this linkage by manually matching previously unmatched patents to Compustat firms using the disambiguated assignee names available in PatentsView. Large-language-model assistance is used to generate candidate matches, which are subsequently verified manually before being incorporated into the final dataset. This extension adds 70,679 patent-to-firm matches over the 1950–2019 period, including 7,147 federally funded patents. Restricting attention to the estimation window (1976–1995), it contributes 17,484 additional matches, of which 2,931 are federally funded. Although these additional observations represent only about 7% of patents in the final estimation sample, they increase the number of matched federally funded patents by more than one-half relative to the original linkage.

Patent citation data combine the historical citation series constructed by Fleming et al. (2019) for patents filed before 1976 with citation records obtained directly from the USPTO for patents filed from 1976 onward. Firm characteristics, including employment, sales, R&D expenditures, industry classification, and firm age, are obtained from Compustat. Finally, I merge three complementary measures of innovation quality. Text-based measures of patent impact are taken from Kelly et al. (2021); patent creativity measures are from Kalyani (2024); and patent-level market value estimates are from Kogan et al. (2017).

4.2 Patent and firm data

The analysis follows patents from the moment inventions are produced to the firms responsible for their underlying research investments. Throughout the paper, patents are assigned to their ultimate corporate owner at the time of filing using the filing-date ownership version of the Dyevre (2023) linkage. This choice is important for identification. The objective is to study how changes in intellectual-property rights affect firms' research decisions when inventions are created, rather than how subsequent mergers, acquisitions, or corporate reorganizations redistribute ownership years later. Assigning patents to filing-date owners therefore aligns observed innovation outcomes with the economic agents that made the original R&D investment and faced the incentives generated by the policy change.

The final matched dataset contains approximately two million patents assigned to Compustat firms over the 1950–2019 period, of which roughly 248,000 are identified as federally funded in Gross and Sampat (2025). The empirical analysis aggregates patents to the firm–technology–year level, where technologies are defined using three-digit Cooperative Patent Classification (CPC) classes.[1] This level of aggregation reduces idiosyncratic patent-level variation while closely approximates the level at which federally funded research projects are organized and selected. Funding agencies award support to research projects within particular technological areas rather than to individual patents, whose number and timing are not known ex ante. Although projects are not directly observed, the firm–technology cell provides the closest observable counterpart to a technological research program within a firm.

Although the analysis focuses on publicly listed firms, this restriction is standard in the innovation literature. Public firms provide consistent longitudinal accounting information over several decades and account for a disproportionate share of U.S. innovative activity, making them particularly well suited for studying how policy-induced changes in intellectual-property rights affect corporate research behavior.

4.3 Sample construction

The ownership classifications in Gross and Sampat (2025) form the basis of the estimation sample. Their archival records distinguish federally funded inventions for which title remained with the government from those for which title was assigned to the contractor subject to a government license. This distinction is central to the analysis because the 1983 reform expanded contractors' ability to retain ownership of federally funded inventions, thereby changing the allocation of intellectual-property rights while leaving the underlying research funding mechanism largely unchanged.

A small fraction of federally funded patents are classified as indeterminate because the archival

[1] Technology classes correspond to the CPC section letter and two-digit class (for example, G02 for optics).

records do not allow ownership status to be assigned unambiguously. For patents filed before 1983, these observations are very few and are excluded following Gross and Sampat (2025). After 1983, however, correspondence with the authors indicates that indeterminate patents assigned to business firms should, for practical purposes, be treated as contractor-owned under the post-reform regime. Accordingly, the baseline analysis classifies these observations as contractor-owned. Reassuringly, only 19 of these patents are assigned to non-firm organizations, and the results are robust to excluding them.

Because universities and nonprofit organizations became subject to a different ownership regime following the Bayh–Dole Act of 1980, I classify assignees by organizational type using the disambiguated PatentsView assignee names. This allows business firms to be distinguished from universities, government agencies, nonprofit organizations, and mixed organizational forms. The baseline sample assigns patents according to filing-date corporate ownership. As a robustness exercise, I exclude patents involving universities, nonprofit organizations, government agencies, or mixed ownership structures to verify that the results are not driven by organizations already operating under different intellectual-property incentives before the 1983 reform or by research projects jointly directed by non-firm actors.

The estimating sample covers patents filed between 1976 and 1995 and is aggregated to the firm–technology–year level. The lower bound is chosen because patent and assignee information become substantially more complete beginning in 1976 with the start of the PatentsView data. Beginning the sample in 1976 also avoids the structural changes that characterized the early 1970s and provides a stable pre-reform period for constructing treatment status and pre-treatment characteristics. The upper bound is chosen to precede the substantial changes in patent eligibility and patenting behavior associated with software and business-method patents during the mid-to-late 1990s, which altered the composition of patented inventions. Restricting the analysis to 1976–1995 therefore maintains a relatively stable institutional and technological environment around the 1983 reform while ensuring consistent data coverage throughout the estimation period.

4.4 Outcome measures

The primary outcomes consider utility patents and capture knowledge spillovers using forward patent citations. Patent citations remain the standard proxy for knowledge diffusion in the innovation literature because they measure the extent to which subsequent inventions build upon earlier discoveries. The baseline specification measures forward cross-technology citations, excluding citations originating from the patent's own technology class. Aggregating these citation flows to the firm–technology–year level captures the extent to which research conducted within a technological area generates knowledge that diffuses beyond its immediate field.

To verify that the estimated effects are not specific to citation-based measures, I also use the text-based innovation measures developed by Kelly et al. (2021), which rely on an entirely different source of variation based on textual similarity across patents. In addition to their publicly available patent-level measures, I use the authors' proprietary patent-pair similarity data for highly similar patent pairs (similarity above 50%). Focusing on these highly similar pairs better aligns the measure with the paper's objective of tracing meaningful knowledge diffusion rather than broad textual overlap and reduces the mechanical tendency of aggregate similarity measures to classify patents from prolific patentees as influential simply because they generate many subsequent patents. These pair-level data allow me to distinguish knowledge reuse within firms from diffusion to external innovators and form the basis for the external diffusion measures analyzed in the paper.

I further examine changes in research direction using the patent creativity measure developed by Kalyani (2024). This measure exploits novel combinations of text bi-grams appearing in patent documents and therefore captures a dimension of innovation that differs from both citations and full-text similarity measures. In the context of this paper, the measure likely reflects both genuine technological novelty and the codification of knowledge into patentable form. Because both processes predate the 1983 reform, the empirical specifications flexibly account for pre-existing differences in creativity across treated and control firm–technology cells. Additionally, although creativity is not used as a primary outcome, it provides an independent source of variation that allows me to assess whether the estimated changes in research direction are consistent across fundamentally different measures of inventive activity.

Finally, I measure the private value of innovation using the patent-level market valuation estimates of Kogan et al. (2017), which infer the economic value of patented inventions from stock-market reactions to patent grants. Together, these complementary outcomes allow the analysis to distinguish whether stronger intellectual-property rights shift innovation toward inventions that are privately more valuable, socially more valuable, or both.

5 Descriptive Evidence

This section documents the empirical relationship between patent ownership arrangements and subsequent knowledge diffusion. The evidence is descriptive and is intended to motivate the identification strategy developed in the following section rather than establish causal effects.

Knowledge diffusion is measured using cross-technology citations. Specifically, for each patent I construct the number of forward citations received from patents outside the focal CPC technology class over the subsequent ten years.[1] Cross-class citations capture the extent to which knowledge generated by an invention diffuses beyond its immediate technological domain and therefore provide a proxy for the broader external relevance of an invention.

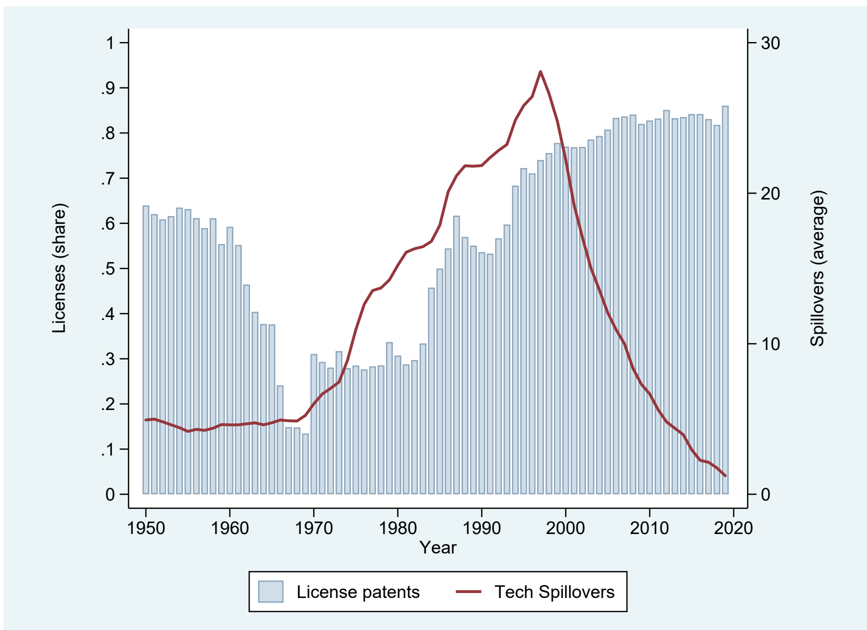


Figure 2: Knowledge Diffusion and Patent Ownership among Federally Funded Patents

Figure 2 plots the evolution of average spillovers among federally funded patents together with the share of patents assigned contractor ownership rights. Two broad patterns emerge. First, contractor ownership becomes substantially less prevalent during the late 1960s and 1970s before rising steadily following the policy changes of the early 1980s. Second, spillovers remain relatively

[1]The ten-year citation window mitigates truncation concerns for more recent patent cohorts, which have had less time to accumulate citations. The results are quantitatively and qualitatively unchanged when spillovers are measured using twenty- or thirty-year forward citation windows.

stable through the 1950s and 1960s, increase sharply from the early 1970s through the late 1990s, and decline for more recent patent cohorts. Importantly, the strongest growth in knowledge diffusion occurs during a period in which contractor ownership remains relatively uncommon and changes little over time. This pattern highlights the difficulty of inferring the effects of ownership arrangements from aggregate time-series variation alone. Changes in knowledge diffusion may reflect a variety of factors, including the cumulative effects of past research investments, evolving technological opportunities, and shifts in the composition of federally funded innovation.

To examine this relationship more directly, I estimate a series of patent-level Poisson regressions relating spillovers to federal funding status and ownership arrangements while progressively introducing richer sets of controls and fixed effects:

$$\begin{aligned} \text{spillovers}_{it} = \beta_0 + \beta_1 \text{Funded}_{it} + \beta_2 \text{License}_{it} + \beta_3 \text{FirmControls}_{it} \\ + \beta_4 \text{PatentControls}_{it} + \sigma_s + \tau_{kt} + \varepsilon_{it}. \end{aligned} \tag{1}$$

The specifications progressively incorporate 2-digit SIC industry, technology-by-filing-year, and funding-agency fixed effects, allowing comparisons among increasingly similar patents. In particular, technology-by-year fixed effects compare patents filed within the same CPC technology class and year, absorbing changes in citation practices and technological opportunities across fields over time. Firm controls include labor productivity, measured as sales per employee, and R&D productivity, measured as R&D expenditures relative to patent output. Patent controls include the patent's total forward citations received within ten years and, for specifications without technology-by-year fixed effects, additionally include CPC technology-class indicators. To account for common shocks and correlation in citation behavior within technological fields over time, standard errors are clustered at the CPC-by-filing-year level.

Table 1 documents three stylized facts. First, federally funded patents generate substantially more cross-technology knowledge diffusion than non-funded patents. Across specifications, federally funded inventions receive approximately 32–43 percent more spillover citations, consistent with a large literature showing that publicly funded research produces large social returns and knowledge spillovers.

Second, conditional on receiving federal funding, patents held under contractor ownership are associated with significantly lower knowledge diffusion. Across the first four specifications, license arrangements are associated with approximately 17–30 percent fewer cross-technology citations.

Table 1: Patent Ownership and Knowledge Diffusion, 1950–2019

	Spillovers						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Government-funded	0.358***	0.367***	0.287***	0.344***			
	(0.044)	(0.050)	(0.041)	(0.047)			
License	-0.246***	-0.257***	-0.162***	-0.214***	-0.115***	-0.102**	-0.062
	(0.047)	(0.054)	(0.045)	(0.050)	(0.041)	(0.046)	(0.049)
N	1990878	1990854	1990220	1990919	25839	24854	23128
Patent Controls	Y	Y	Y	Y	Y	Y	Y
Firm Controls	Y	Y	Y	Y	Y	Y	Y
Year FE	✓	✓			✓		
CPC FE	✓	✓			✓		
Industry FE		✓		✓			
CPC-Year FE			✓	✓		✓	
Agency FE					✓	✓	
Agency-CPC-Year FE							✓
SE	cpc-year	cpc-year	cpc-year	cpc-year	cpc-year	cpc-year	cpc-year

Notes: The unit of observation is the patent. The dependent variable is the number of cross-CPC forward citations received within ten years of the filing year. All specifications are estimated using Poisson pseudo-maximum likelihood (PPML). Firm controls include labor productivity (sales per employee) and R&D productivity (R&D expenditures divided by patents filed). Patent controls include total forward citations received within ten years. Standard errors are clustered at the CPC-by-filing-year level. Year fixed effects refer to filing year. (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$)

The estimated relationship remains remarkably stable as increasingly demanding industry and technology fixed effects are introduced, indicating that it is not driven simply by differences in citation intensity across technological fields or industries.

Finally, restricting the sample to federally funded patents with identifiable funding agencies yields smaller, though generally still statistically significant, estimates. The attenuation reflects both the substantial reduction of the sample and the introduction of agency fixed effects, suggesting that part of the raw association may be attributable to differences in the types of projects funded by different agencies. Nevertheless, the negative relationship between contractor ownership and subsequent knowledge diffusion remains evident throughout. Appendix Table C.1 further shows that this pattern is robust to excluding self-citations from the spillover measure. If anything, the negative association between contractor ownership and spillovers becomes somewhat stronger in specifications that include agency controls approaching in magnitude the rest estimates.

Taken together, these descriptive findings suggest that although federally funded research generates broad knowledge spillovers, assigning contractor ownership rights is associated with substantially narrower diffusion of those inventions. These patterns, however, should not be interpreted causally. Funding agencies may systematically assign different ownership arrangements to different firms, technologies, or research projects, and these underlying differences may themselves influence subsequent knowledge diffusion. The following section therefore develops an identification strategy that exploits the 1983 reform in government patent policy to isolate the causal effect of contractor ownership on knowledge diffusion.

6 Identification Strategy

The objective is to estimate the causal effect of stronger contractor ownership rights over federally funded inventions on subsequent innovation outcomes. The central empirical challenge is that patent ownership is not randomly assigned. Firms performing federally funded research differ systematically from other firms, and federal agencies may direct funding toward projects with particular technological, scientific, or commercial characteristics. I address these concerns by exploiting the 1983 Reagan Memorandum on Government Patent Policy, which established a government-wide presumption that contractors should retain title to inventions arising from federally funded research.

Identification comes from a difference-in-differences design implemented at the firm–technology–year level, where a technology is defined by a CPC class. Federal research support is allocated to specific technological programs rather than uniformly across firms. Consequently, exposure to the ownership reform varies across technologies within the same firm, making the firm–technology cell the appropriate unit of analysis.

Accordingly, a firm–technology cell is classified as treated if the firm received at least one federally funded patent in that technology before 1983. The comparison group consists of firm–technology cells belonging to firms that never receive federal funding during the sample period, while firm–technology cells first exposed to federal funding after 1983 are excluded from the estimation sample. Treatment status is therefore determined entirely by pre-reform exposure to federally funded research. Using never-treated controls ensures that the comparison group does not experience a contemporaneous change in the degree of appropriability during the analysis period.

The baseline estimating equation is

$$Y_{fkt} = \beta\,(Treated_{fk} \times Post_t) + \phi_{fk} + \tau_{kt} + \sigma_{st} + X_{fkt} + \varepsilon_{fkt}, \tag{2}$$

where Y_{fkt} denotes an innovation outcome for firm f, technology k, and year t, $Treated_{fk}$ indicates pre-1983 exposure to federally funded research, and $Post_t$ equals one after the 1983 reform. The specification includes firm–technology fixed effects, ϕ_{fk}, CPC-by-year fixed effects, τ_{kt}, and SIC-2-by-year fixed effects, σ_{st}. The coefficient of interest, β, identifies the average effect of the reform on previously federally funded firm–technology cells relative to comparable never-funded cells. The vector X_{fkt} contains flexible time paths constructed by interacting pre-treatment quartile indicators for firm size, R&D intensity, technological breadth, patenting activity, innovation quality, and firm age with year fixed effects. Standard errors are clustered at the firm level, allowing for arbitrary correlation in outcomes across technologies and years within firms.

The identifying assumption is that, absent the 1983 change in ownership rules, treated and comparison firm–technology cells would have followed parallel trends conditional on the included fixed effects and controls. Two threats to this assumption merit particular attention.

The first concern is selection into federally funded research. Firms receiving federal support differ systematically from never-funded firms in characteristics that may also influence subsequent innovation outcomes. I mitigate this concern through both sample construction and regression adjustment. First, I construct the estimation sample using a two-step matching procedure. Treated firm–technology cells are matched exactly within CPC technology classes and two-digit SIC industries, after which propensity score matching is performed using pre-treatment firm characteristics and innovation outcomes. The resulting matched sample exhibits substantial common support and substantially improved balance in pre-treatment characteristics, as documented in the next section[1]. Second, the regression specification allows firms with different initial characteristics to follow distinct post-1983 trajectories by interacting pre-treatment characteristics with year fixed effects.

The second concern is that the early 1980s coincided with broader changes in U.S. innovation policy, including the implementation of Bayh–Dole and a general strengthening of patent rights. The research design mitigates this concern by exploiting differential exposure to a policy that specifically altered ownership rights over federally funded inventions. CPC-by-year fixed effects absorb shocks common to technologies, while SIC-2-by-year fixed effects absorb contemporaneous industry-level shocks. Identification therefore comes from comparing treated and comparison firm–

[1] for evidence on common support see Appendix figure A1

technology cells operating in similar technological and industrial environments at the same point in time.

The timing of the reform further supports its interpretation as a discrete policy shock rather than the continuation of a widely anticipated trend. In June 1980, the Senate rejected by a vote of 60–34 an amendment that would have extended Bayh–Dole's ownership provisions to large federal contractors.[1] The subsequent expansion of contractor ownership through executive action therefore followed a policy path that had previously been rejected through the legislative process, reducing concerns that firms anticipated the reform well in advance. Consistent with this interpretation, the descriptive evidence presented in Section 5 shows little indication of anticipatory changes in either federally funded patenting or contractor ownership before 1983.

The next section evaluates the identifying assumptions directly. I first examine covariate balance after matching and then estimate event-study specifications to assess whether treated and comparison firm–technology cells followed similar pre-treatment trends. Together, these exercises provide direct evidence on the credibility of the identifying assumptions underlying the empirical design.

7 Validation of the Research Design

This section evaluates the empirical support for the identifying assumptions underlying the difference-in-differences design. I present three complementary pieces of evidence. First, I examine whether the matching procedure produces comparable treated and control firm–technology cells. Second, I assess whether the two groups exhibit similar pre-treatment spillover dynamics. Finally, I estimate an event-study specification to evaluate whether any differential trends emerge before the 1983 reform.

7.1 Covariate Balance

Table 2 reports covariate balance between treated and matched control firm–technology cells. The matching procedure achieves close balance for the variables most directly related to the identifying assumption. In particular, treated and control observations exhibit nearly identical pre-treatment

[1] See Letter from Sen. Gaylord Nelson, Chairman, Senate Select Committee on Small Business, to Rep. Robert W. Kastenmeier, Chairman, Subcommittee on Courts, Civil Liberties, and the Administration of Justice of the House Committee on the Judiciary (June 1980), reprinted in *House Hearings on President's Industrial Innovation Program*, pp. 877–878. I became aware of this source through Eisenberg (1996).

spillover levels and highly comparable spillover trends. Productivity, R&D intensity, and pre-treatment trends in employment, patent creativity, and technological breadth are likewise well balanced.

Table 2: Covariate Balance

	Levels				**Trends**			
	Treated Firms	Matched Controls	Non-treated Firms	Std. Diff.	Treated Firms	Matched Controls	Non-treated Firms	Std. Diff.
Spillovers	15.40	15.29	16.37	0.005	0.52	0.19	-0.08	0.005
	(12.54)	(22.08)	(23.77)		(7.01)	(5.61)	(16.08)	
# patents filed	9.78	3.43	1.89	0.561	0.006	-0.40	-0.02	0.227
	(14.93)	(5.69)	(2.39)		(1.93)	(1.66)	(0.61)	
Patent creativity	1.15	1.08	1.09	0.127	0.006	0.001	-0.02	0.020
	(0.41)	(0.73)	(0.71)		(0.23)	(0.28)	(0.36)	
Employees	85.43	32.02	23.32	0.518	0.36	0.34	0.27	0.003
	(128.17)	(71.19)	(82.87)		(6.89)	(4.41)	(2.18)	
TFP	121.18	118.88	106.80	0.017	0.87	0.85	0.35	0.003
	(135.05)	(120.29)	(136.02)		(9.60)	(9.04)	(10.29)	
R&D intensity	23.52	13.34	6.30	0.051	0.88	0.76	0.27	0.018
	(46.14)	(29.87)	(32.22)		(5.24)	(8.41)	(3.16)	
Age (in 1983)	3.17	4.35	4.50	-0.262	—	—	—	—
	(2.74)	(5.72)	(5.53)					
# technology classes	24.26	5.66	4.66	1.917	-0.07	-0.02	0.12	-0.053
	(13.15)	(3.91)	(4.53)		(1.35)	(0.87)	(0.89)	
N	1,404	1,122	14,008		1,404	1,122	14,008	

Notes: The table reports pre-treatment means for treated firm–technology cells, matched control firm–technology cells, and all non-treated firm–technology cells. Standard deviations are reported in parentheses. Standardized differences compare treated and matched control observations. Patent creativity is measured using the creativity index developed by Kalyani (2024). Employees are reported in thousands and R&D intensity is R&D expenditure (in millions of dollars) per CPC class. Firm age is measured as the number of years since the earliest year the firm can be identified in the data, defined as the minimum of its IPO year, first patent filing year, and first appearance in Compustat. The number of technology classes corresponds to the number of distinct CPC classes in which a firm filed patents during a given year. Covariates with standardized differences exceeding 0.10 enter the baseline specification through interactions between pre-treatment quartile indicators and filing-year fixed effects.

Matching does not, however, eliminate all observable differences. Treated firms remain larger, patent more intensively, operate across a broader range of technologies, and differ somewhat in baseline patent creativity and firm age. Rather than requiring treated and control observations to evolve similarly along these dimensions, the baseline specification explicitly accommodates differential dynamics by interacting filing-year fixed effects with pre-treatment quartile bins of all covariates whose standardized differences exceed 0.10. This approach allows initially different firms to follow different trajectories over time while identifying the treatment effect from deviations beyond those flexible trends.

7.2 Graphical Evidence on Pre-Treatment Dynamics

A key identifying assumption is that, absent the 1983 reform, treated and control firm–technology cells would have followed similar spillover trajectories.

Figure 3 plots average spillovers for treated and matched control observations over the 1976–1995 estimation window. The two groups display similar pre-treatment dynamics despite modest differences in levels, with no indication of systematic divergence before the reform. Their trajectories begin to separate only after the change in patent ownership policy. Appendix Figure A2 presents the corresponding series for the full 1950–2019 period and shows a similar pattern.

Figure 4 presents the same comparison after partialling out the fixed effects and flexible control paths included in the baseline specification. This figure provides the closest graphical analogue to the identifying assumption underlying the estimating equation. After residualization, treated and control observations track one another closely throughout the pre-treatment period, while a sustained decline in spillovers among treated firm–technology cells emerges only after 1983.

The similarity of the residualized pre-treatment trajectories is particularly informative given the remaining differences in observable firm characteristics documented in Table 2. The absence of differential pre-treatment dynamics suggests that the subsequent decline in spillovers is unlikely to reflect the continuation of pre-existing trends.

7.3 Event-Study Evidence

I next estimate a dynamic event-study specification that replaces the post-treatment indicator with a full set of leads and lags relative to the 1983 reform. This specification allows the evolution of treatment effects before and after the policy change to be examined directly.

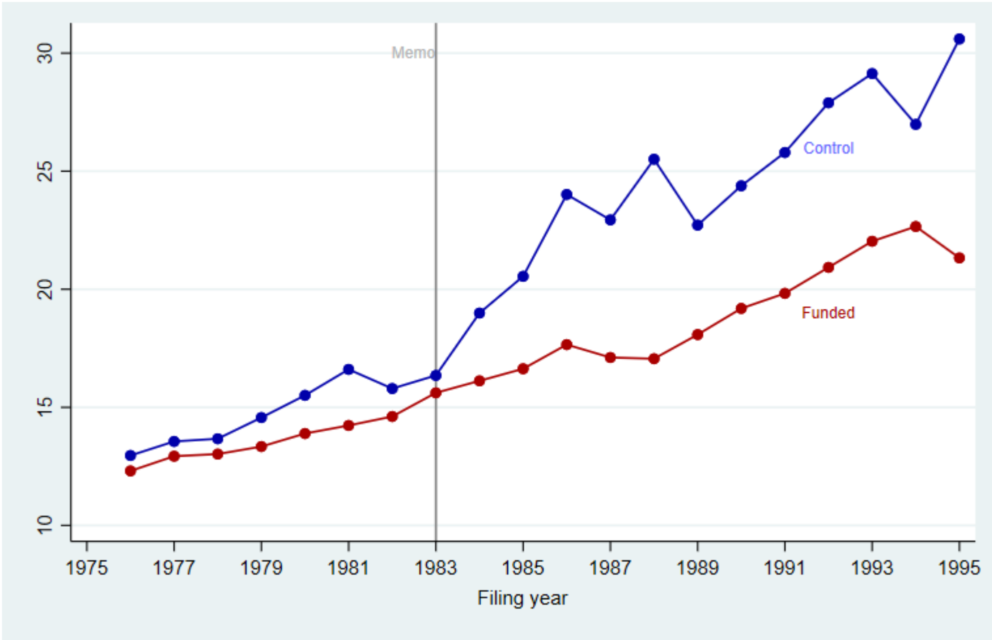


Figure 3: Spillovers by Matched Group, 1976–1995: Raw Comparison

Notes: The figure plots average spillovers for treated and matched control firm–technology cells during the estimation period. Spillovers are measured as cross-technology forward citations received over the subsequent ten years. The vertical line marks the 1983 reform.

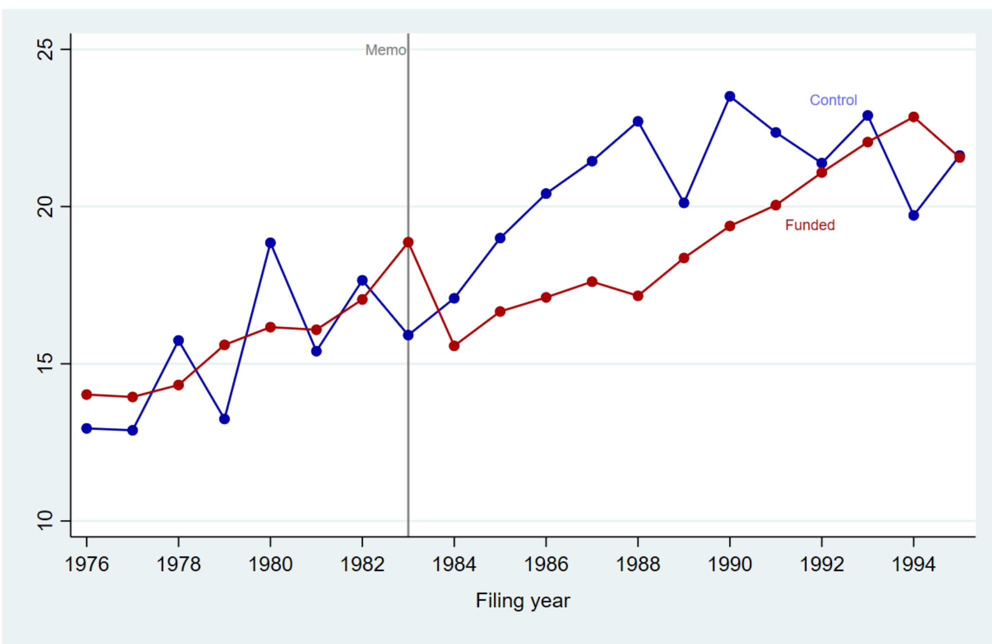


Figure 4: Spillovers by Matched Group, 1976–1995: Baseline Specification

Notes: The figure plots residualized spillovers after partialling out the firm–technology, technology–year, and industry–year fixed effects together with the flexible time paths included in the baseline specification. Spillovers are measured as cross-technology forward citations received over the subsequent ten years.

Figure 5 reports the estimated event-study coefficients. The pre-treatment estimates fluctuate around zero and are individually statistically indistinguishable from zero, providing no evidence of systematic differential dynamics prior to the reform. Following the reform, the estimated effects become negative and remain persistently below zero throughout the post-treatment period. The timing and persistence of these estimates closely mirror the patterns observed in the graphical evidence and are consistent with the baseline difference-in-differences results.

Taken together, the balance diagnostics, graphical evidence, and event-study estimates provide consistent support for the research design. While treated and control firms differ along some observable dimensions, those differences are accommodated through flexible time paths in the baseline specification. After accounting for these differences, treated and control firm–technology cells exhibit remarkably similar spillover dynamics prior to the 1983 reform, with the divergence emerging only after the change in patent ownership policy.

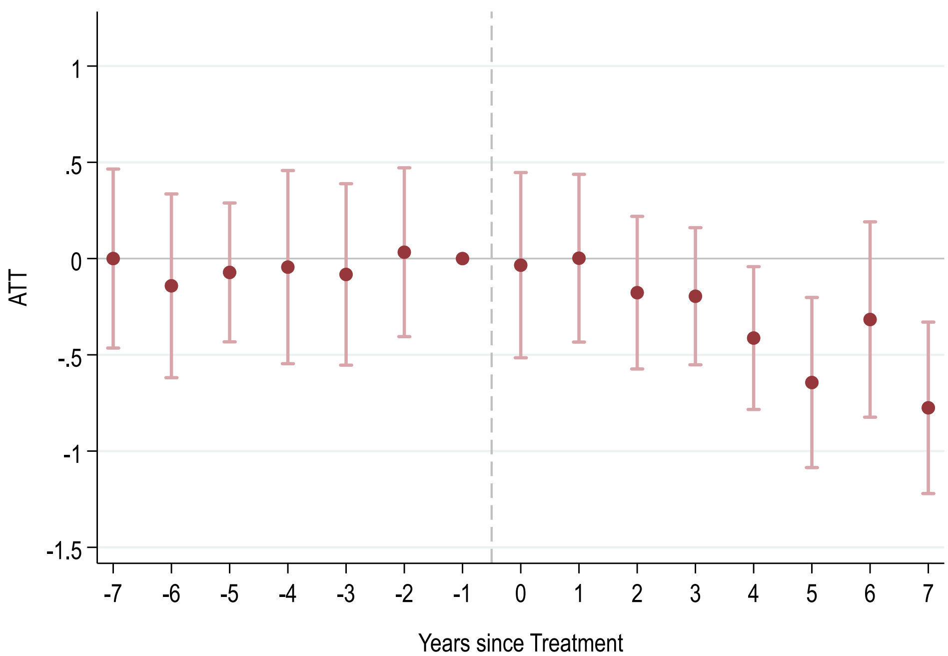


Figure 5: Event-Study Estimates

Notes: The figure reports dynamic treatment effects relative to the 1983 reform. Vertical bars indicate 95% confidence intervals based on standard errors clustered at the firm level. Estimates are obtained from the baseline Poisson specification including firm–technology, technology–year, and industry–year fixed effects together with flexible time paths for imbalanced pre-treatment covariates. The year before the policy change is omitted (year -1).

8 Main Results

This section presents the main evidence on the consequences of the 1983 ownership reform for the diffusion and value of federally funded inventions. I begin by examining its effect on cross-technology knowledge spillovers. I then investigate whether the decline reflects a broader reduction in technological significance using text-based measures of follow-on innovation from Kelly et al. (2021). Finally, I examine whether the reduction in external diffusion was accompanied by changes in the private market value of treated inventions.

8.1 Effects on Cross-Technology Spillovers

The conceptual framework developed in the previous section predicts that stronger contractor ownership affects innovation through two complementary channels. First, it increases firms' ability to appropriate the returns from federally funded inventions. Second, by changing the relative profitability of different research projects, it shifts innovation toward inventions with greater private appropriability but potentially narrower technological scope. The empirical analysis begins by examining whether these changes are reflected in cross-technology knowledge spillovers.

Table 3 reports the baseline difference-in-differences estimates. All specifications include firm–technology fixed effects, technology-by-year fixed effects, and industry-by-year fixed effects. The columns sequentially introduce flexible time paths based on pre-treatment firm and innovation characteristics to allow treated and control observations with different initial conditions to follow different underlying trajectories over time.

The specification including only fixed effects produces a small and statistically insignificant estimate. Once I allow firms with different pre-treatment innovation trajectories to evolve flexibly over time by interacting baseline patent creativity with year fixed effects, the estimated treatment effect becomes substantially larger, negative, and statistically significant. The estimated coefficient remains remarkably stable as additional flexible time paths are introduced for firm size, patenting activity, R&D expenditures, technological breadth, and firm age.

In the preferred specification, contractor ownership reduces cross-technology spillovers by 22 percent.[1] This is a quantitatively meaningful decline given that cross-technology citations capture knowledge flows extending beyond the technological area in which the invention originated. The stability of the estimates across specifications further suggests that the results are unlikely to reflect

[1] For Poisson models, percentage effects are calculated as $\exp(\beta) - 1$.

Table 3: Effects on Cross-Technology Spillovers

	Spillovers						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated x Post	-0.107	-0.287***	-0.284***	-0.265***	-0.260***	-0.240**	-0.250***
	(0.107)	(0.085)	(0.090)	(0.093)	(0.088)	(0.095)	(0.088)
Industry-Year FE	✓	✓	✓	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓	✓	✓	✓
Patent Creativity		✓	✓	✓	✓	✓	✓
Employees			✓	✓	✓	✓	✓
Patents filed				✓	✓	✓	✓
R&D					✓	✓	✓
Agegroup						✓	✓
# of CPCs							✓
SE	firm	firm	firm	firm	firm	firm	firm
N	24,911	19,263	19,222	19,222	18,466	18,466	18,466

Notes: The dependent variable is the number of forward citations received from patents outside the focal CPC technology class within ten years of the patent filing year. All specifications are estimated using Poisson pseudo-maximum likelihood (PPML). Control variables enter as interactions between pre-treatment quartile indicators and year fixed effects, allowing for flexible differential time paths across baseline covariate groups. Patent creativity is measured using the index developed by Kalyani (2024). Standard errors are clustered at the firm level. (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$)

differential pre-existing trends in observable innovation capacity or firm characteristics.

The event-study estimates presented in Figure 5 reinforce this interpretation. As discussed in Section 7, treated and control firm–technology cells exhibit parallel trends throughout the pre-reform period, while the decline in cross-technology spillovers emerges only after the 1983 policy change. Table 4 summarizes three additional exercises that sharpen the interpretation of the main result.

First, excluding patents jointly assigned to non-firm organizations increases estimates by about 25 percent. This restriction yields a cleaner measure of the technological influence of inventions reflecting firms' own research decisions. Second, restricting spillovers to citations originating from other firms strengthens the estimated effect by almost 50 percent[1]. Together, these findings indicate that the decline in technological influence is concentrated in knowledge that diffuses beyond the originating firm and is robust to alternative definitions of external spillovers.

[1] in Appendix, tables with complete progression of specifications are available (Tables A2 - A4)

Table 4: Effects on Cross-Technology Spillovers: Alternative Definitions

	Spillovers			
	Cross-CPC	Cross-CPC firms only	Cross-CPC&firm	Cross-CPC&firm firms only
Treated x Post	-0.250***	-0.314***	-0.366***	-0.372***
	(0.088)	(0.112)	(0.107)	(0.105)

Notes: The table reports estimates from the fully saturated specification (column 7 of Table 3). The baseline outcome counts forward citations originating from patents outside the focal CPC technology class. The second column further excludes citations originating from patents with non-firm co-assignees. The third column restricts spillovers to forward citations that are both outside the focal CPC technology class and outside the originating firm (excluding self-citations). The fourth column applies both restrictions simultaneously by excluding citations from non-firm entities and from the originating firm. All specifications include the full set of fixed effects and controls from the baseline specification. Standard errors are clustered at the firm level. (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$)

Although cross-technology citations provide a natural measure of knowledge spillovers, citation counts may also reflect changes in citation practices or in firms' propensity to build on their own prior inventions. The next subsection therefore examines whether the decline in spillovers is also evident in text-based measures of technological significance that are independent of citation behavior.

8.2 Text-Based Evidence on Technological Significance

I next examine whether the decline in spillovers extends beyond citation-based measures by using the text-similarity approach developed by Kelly et al. (2021). Rather than relying on patent citations, this measure captures technological influence directly through the textual similarity of patents rather than through citation behavior.

The publicly available patent-level measure provided by Kelly et al. (2021) produces negative but statistically insignificant estimates in this setting, reported in Appendix Table A5. This outcome is not unexpected. The public measure aggregates all technologically similar inventions, regardless of whether they are developed by the originating firm or by external innovators. As a result, it combines broad technological advances with subsequent within-firm development and refinement, making it less informative about whether stronger ownership rights altered the external diffusion and direction of innovative activity.

To better align the outcome with the mechanism studied here, I reconstruct the Kelly measure using a dataset generously provided by the authors containing the underlying patent-pair similarity indices for patent pairs with text similarity exceeding 50 percent. Following Kelly et al. (2021), I reconstruct their measure of patent significance by computing forward and backward similarity indices and aggregating them to the firm–technology–year level. Unlike the publicly available measure, these micro-level data allow me to identify patent-pair ownership within my matched sample and distinguish between within-firm similarity and similarity extending to other firms. I begin by including all patent pairs, including those assigned to the same firm. This specification most closely matches the original Kelly measure within my estimation sample and therefore serves as the baseline.

Table 5: Effects on Text-Based Technological Significance

	Patent Significance				
	(1)	(2)	(3)	(4)	(5)
Treated x Post	-0.096**	-0.132**	-0.110**	-0.108**	-0.096*
	(0.046)	(0.052)	(0.052)	(0.049)	(0.051)
Industry-Year FE	✓	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓	✓
Patent Creativity		✓	✓	✓	✓
Employees			✓	✓	✓
Citations within CPC				✓	✓
R&D					✓
SE	firm	firm	firm	firm	firm
N	24,634	18,727	18,690	18,435	17,782

Notes: The dependent variable is the reconstructed KPST measure based on patent pairs with text similarity exceeding 50 percent, following Kelly et al. (2021). All similarity pairs, including those assigned to the same firm, are included. Standard errors are clustered at the firm level.
(* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$)

Table 5 shows that the reconstructed measure declines significantly following the reform. Across specifications, the preferred estimates imply a reduction in technological significance of approximately 9–13 percent. In contrast to the publicly available patent-level measure reported in Appendix Table A5, the reconstructed measure yields economically meaningful and statistically significant estimates throughout. This finding indicates that the decline in spillovers is not specific to citation-based outcomes but extends to text-based measures of technological influence.

The reconstructed measure, however, still combines internal and external innovation. The Kelly measure captures the extent to which subsequent inventions remain technologically similar to earlier work, but textual similarity alone cannot distinguish broad technological advances from increasingly specialized within-firm refinement. This is central to the current analysis since a shift in the orientation of innovation to more appropriable but narrower projects could be muted in such a measure. A testable way to separate those two aspects is to focus on the external diffusion of innovation. Although incremental product development is important for a firm, such an innovation is much more likely to remain within firm whereas genuinely new technological trajectories are more likely to generate follow-on inventions by other organizations. The testable implication is that if the decline in technological significance primarily reflects a reduction in innovation with external technological relevance, the estimated effect should become more pronounced once within-firm innovation is excluded. The next subsection examines this prediction directly.

8.3 External Technological Significance

To isolate the external component of technological influence, I reconstruct the Kelly measure after excluding patent pairs in which the patent is assigned to the same identified firm. The resulting outcome focuses on technologically similar inventions developed by other firms and therefore provides a direct measure of external technological significance.

Table 6 shows that excluding same-firm patent pairs substantially strengthens the estimated effect. Depending on the specification, external technological significance declines by roughly 20–35 percent following the reform, considerably larger than the 9–13 percent decline obtained when within-firm follow-on innovation is retained.

A notable feature of the results is the remarkable consistency across fundamentally different measures of technological influence. The within-firm component attenuates both citation-based spillovers and text-based technological similarity. Although the different types of innovation within the firm cannot be identified separately, this evidence provides an indication that innovation relevant primarily to the firm only is at play. More directly, the much larger estimates obtained after excluding same-firm patent pairs indicate that the contraction is concentrated in innovation that is externally relevant. Thus, direct and indirect evidence point to a pattern which is consistent with the conceptual framework.

If stronger contractor ownership shifted innovative effort toward more privately appropriable

Table 6: Effects on External Text-Based Technological Significance

	Patent Significance						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated x Post	-0.320***	-0.355***	-0.271**	-0.257**	-0.243*	-0.208	-0.219*
	(0.108)	(0.115)	(0.137)	(0.130)	(0.133)	(0.128)	(0.128)
Industry-Year FE	✓	✓	✓	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓	✓	✓	✓
Patent Creativity		✓	✓	✓	✓	✓	✓
Employees			✓	✓	✓	✓	✓
Patents filed				✓	✓	✓	✓
R&D					✓	✓	✓
# of CPCs						✓	✓
Agegroup							✓
SE	firm	firm	firm	firm	firm	firm	firm
N	25,266	19,288	19,248	19,248	18,532	17,872	17,872

Notes: The specification is identical to Table 5, except that patent pairs assigned to the same firm are excluded when constructing the dependent variable.
(* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$)

research, subsequent within-firm development may remain textually similar to the originating invention even when the broader technological scope of that research narrows. Because the Kelly measure is based on textual similarity, such within-firm follow-on innovation continues to contribute to measured technological significance, partially masking the decline observed in the external component.

The next subsection gets a step closer to isolating the external component of the Kelly measure by providing a clearer metric of diffusion while addressing a remaining limitation of the data.

8.4 Allocation of Follow-on Innovation

The previous subsection showed that the decline in technological significance is considerably stronger once within-firm innovation is excluded. A remaining limitation, however, is that the Kelly measure combines two distinct margins. Because it is constructed as the difference between forward similarity (*"impact"*) and backward similarity (*"novelty"*), the larger decline in external significance could reflect weaker technological impact, greater similarity to existing technologies, or both.

To isolate the diffusion margin more directly, I next examine the external diffusion share, defined as the fraction of forward technological similarity generated by patents assigned to other firms. Formally, the measure is the ratio of cross-firm to total forward similarity accumulated over subsequent years. Unlike the Kelly measure, it is based exclusively on forward similarity and therefore abstracts from the novelty component. It directly captures how technologically similar follow-on innovation is allocated between the originating firm and external innovators.

The sample available has two remaining limitations for my analysis. First, the underlying patent-pair data retain only observations with text similarity exceeding 50 percent, a very selective part of the full similarity distribution[1]. Second, as a sum of similarity indexes, the measure assigns greater weight to within-firm pairs, which can approach 100 percent similarity, than to cross-firm pairs, whose similarity is naturally closer to the 50 percent threshold. Because only a third of the patents in the relevant period are matched in my sample, some within-firm pairs likely remain in the measure after excluding the identified ones. To address both limitations, I replace the sum of similarity indexes with the count of patent pairs in this highly selective dataset. The count of highly similar external pairs provides a natural measure of the breadth of external diffusion, assigning equal weight to each patent pair and focusing on the outcome of interest: how often a patent gives rise to technologically similar inventions outside the originating firm.

Table 7 reports a large and robust decline in the external diffusion share following the reform. Across specifications, the preferred estimates imply that the share of technologically similar follow-on innovation generated outside the originating firm falls by approximately 18–20 percent. The estimates remain remarkably stable as flexible controls are sequentially introduced.

Figure 6 further shows that the decline emerges only after the 1983 reform and is not preceded by differential pre-treatment trends.

[1] Kelly et al. (2021) report that the average similarity score in their data is 10.2 percent, while the 95th percentile is only 22.9 percent. Patent pairs with similarity exceeding 50 percent therefore represent an extreme upper tail of the similarity distribution

Table 7: Effects on the External Diffusion Share

	Externalization Share						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated x Post	-0.123***	-0.171***	-0.190***	-0.195***	-0.202***	-0.183***	-0.189***
	(0.038)	(0.042)	(0.044)	(0.042)	(0.041)	(0.041)	(0.041)
Industry-Year FE	✓	✓	✓	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓	✓	✓	✓
Patent Creativity		✓	✓	✓	✓	✓	✓
Employees			✓	✓	✓	✓	✓
Patents filed				✓	✓	✓	✓
R&D					✓	✓	✓
# of CPCs						✓	✓
Agegroup							✓
SE	firm	firm	firm	firm	firm	firm	firm
N	24,023	18,479	18,447	18,447	17,909	17,374	17,374

Notes: The dependent variable is the external diffusion share, defined as the share of high-similarity forward patent pairs assigned to other firms. All specifications are identical to those in Table 6. Standard errors are clustered at the firm level. (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$)

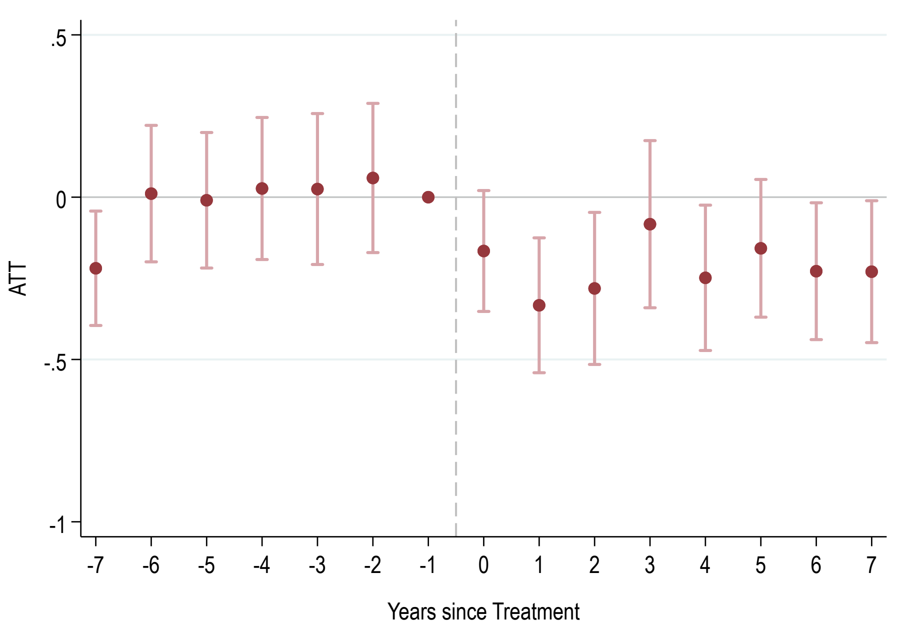


Figure 6: Event-Study Estimates: External Diffusion Share

Notes: Same specification as the previous event-study figure, with the external diffusion share as the dependent variable. Vertical bars denote 95% confidence intervals. The year before the policy change is omitted (year -1).

The conclusions are robust to several alternative definitions of external diffusion. Appendix Tables A9–A13 first show that the results are virtually unchanged when the external diffusion share is constructed using similarity weights rather than counts. More importantly, I redefine external diffusion as technologically similar follow-on innovation occurring outside both the originating firm and the focal CPC technology class, mirroring the citation-based spillover measure used throughout the paper. The estimated effects remain quantitatively similar under this more restrictive definition. Finally, excluding patents jointly assigned with universities and other non-business organizations strengthens the results somewhat. Across all alternative definitions, the estimated effects remain negative, statistically significant at the one-percent level, and economically similar to the baseline.

Taken together, the Kelly-based evidence paints a consistent picture. The reform reduced the external technological significance of innovation produced in federally funded firm-technology units while lowering the share of technologically similar follow-on innovation occurring outside the originating firm. Unlike the previous Kelly measures, which combine technological significance and novelty, the external diffusion share isolates the destination of follow-on innovation, providing the clearest evidence of a systematic contraction in the external technological relevance of federally funded units.

8.5 Private Market Value

The preceding results show that stronger contractor ownership reduced the external technological reach of funded firm-technologies. Whether this decline reflects a reduction in the overall value of innovation or a reallocation of returns from social to private value remains an open question. To examine this issue, I use the patent-level market value measure developed by Kogan et al. (2017), aggregated to the firm–technology–year level. Because patent grants, rather than patent applications, constitute the information event observed by financial markets, market-value measures are matched using grant year.[1]

Table 8 reports a positive and economically meaningful effect of the reform on patent market value. Once flexible time paths are introduced for pre-treatment patent creativity, market capitalization, firm size, citation intensity within the CPC class, stock-market volatility, and R&D, the estimated effect becomes positive, stable, and statistically significant. The preferred estimates imply an increase in private patent value of approximately 30–40 percent.

[1] Before 1999, patent applications were generally not published until grant. With the *American Inventors Protection Act* (AIPA) in 1999, most applications have been published 18 months after filing.

Table 8: Effects on Patent Market Value

	Patent Market Value						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated x Post	0.121	0.208*	0.261**	0.273**	0.276**	0.333**	0.290*
	(0.126)	(0.116)	(0.117)	(0.139)	(0.139)	(0.137)	(0.160)
Full FE	✓	✓	✓	✓	✓	✓	✓
Patent Creativity		✓	✓	✓	✓	✓	✓
Market Capitalization			✓	✓	✓	✓	✓
Employees				✓	✓	✓	✓
Citations within CPC					✓	✓	✓
Stochastic Volatility						✓	✓
R&D							✓
SE	firm	firm	firm	firm	firm	firm	firm
N	20,132	15,254	15,092	15,080	14,841	14,712	14,290

Notes: Outcome variable is the patent market-value measure of Kogan et al. (2017), aggregated to the firm–technology–year level. Market values are matched by grant year. Stochastic volatility is calculated as the residual from regressions of excess stock returns on market returns and the Fama–French three factors using CRSP daily data. Market capitalization equals shares outstanding multiplied by the stock price at the end of the fiscal year. All specifications include firm–technology, technology×year, and SIC2×year fixed effects. Standard errors are clustered at the firm level. (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$)

The event-study estimates in Figure 7 show no evidence of differential pre-treatment trends and indicate that the increase in market value emerges only after the 1983 reform.

Taken together, the evidence points to a consistent pattern. Stronger contractor ownership reduced the external technological significance and diffusion of federally funded inventions while increasing their private market value. This combination of results is consistent with the conceptual framework developed in Section 2: by increasing the returns to private appropriation, the reform shifted innovative effort toward inventions generating greater private value but narrower external technological relevance. The evidence therefore suggests that allocating ownership rights over publicly funded research induces a trade-off, affecting not only the distribution of returns from innovation but also the direction of innovative effort itself.

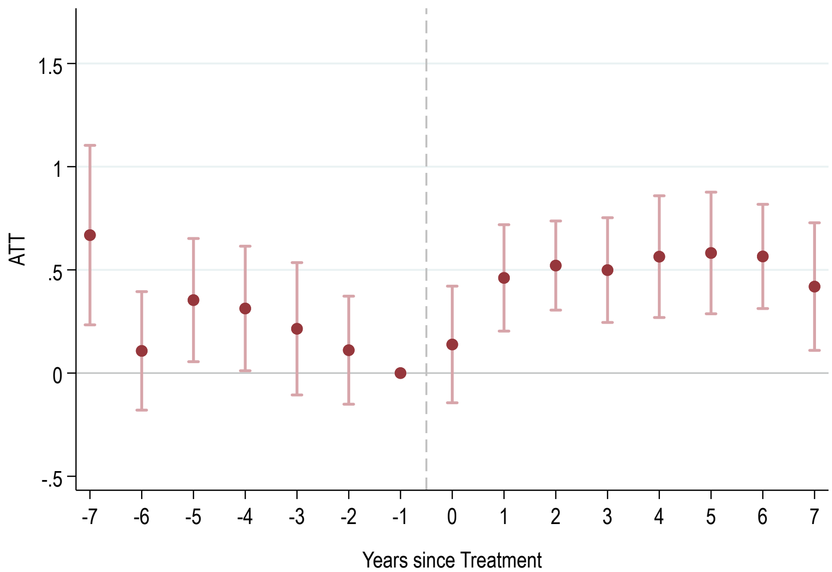


Figure 7: Event-Study Estimates: Patent Market Value

Notes: Same specification as the previous event-study figure, with patent market value as the dependent variable. Flexible time paths are additionally included for pre-treatment market capitalization and idiosyncratic stock-return volatility. Vertical bars denote 95% confidence intervals.

9 Mechanisms

The previous section established that expanding contractor ownership reduced knowledge spillovers while increasing the private market value of federally funded inventions. By strengthening firms' ability to appropriate the returns from publicly funded research, the reform increased the relative profitability of projects whose benefits could be more fully internalized. A natural implication of this trade-off is that such a change need not reduce innovative activity overall. Instead, a reallocation of innovative effort toward inventions with greater private value but narrower external technological influence can take place.

This section examines that mechanism. I first investigate whether the decline in spillovers is concentrated among the most influential inventions. I then examine which firms account for these changes and whether the observed reallocation is accompanied by corresponding changes in patent market value, technological significance, and inventive novelty.

9.1 Which Inventions Disappear?

The average decline in spillovers documented in the previous section does not imply that all inventions were affected equally. If stronger contractor ownership changes the composition of research rather than uniformly reducing knowledge diffusion, the largest effects should be concentrated among inventions that historically generated the greatest external technological influence. To examine this prediction, Table 9 estimates the baseline specification separately for patents belonging to different quartiles of the cross-technology citation distribution.

Table 9: Spillovers Across the Distribution

		Spillovers				
A: 0-25%	Treated x Post	0.186	0.264*	0.255*	0.230	-0.074
		(0.120)	(0.137)	(0.140)	(0.143)	(0.168)
	N	14264	13178	13173	13038	14162
B: 25-50%	Treated x Post	-0.115	0.015	-0.046	-0.032	-0.074
		(0.119)	(0.115)	(0.132)	(0.130)	(0.168)
	N	18666	17005	16982	16778	14162
C: 50-75%	Treated x Post	-0.156*	-0.114	-0.087	-0.106	-0.074
		(0.084)	(0.094)	(0.100)	(0.097)	(0.168)
	N	18345	16784	16762	16595	14162
D: 75-100%	Treated x Post	-0.221	-0.521***	-0.480***	-0.435***	-0.074
		(0.167)	(0.135)	(0.141)	(0.139)	(0.168)
	N	18586	17054	17013	16881	14162
Full FE		✓	✓	✓	✓	✓
Patent Creativity			✓	✓	✓	✓
Employees				✓	✓	✓
Citations, within cpc					✓	✓
R&D Expenditure						✓

Notes: The dependent variable is cross-CPC forward citations excluding citations to patents assigned to the same firm. Estimates are reported separately for the pre-treatment spillover quartiles. Full FE refers to firm–technology, technology–year, and two-digit SIC industry–year fixed effects. Standard errors are clustered at the firm level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The estimates reveal that the decline in spillovers is highly uneven across the distribution. Before introducing flexible controls for firms' pre-treatment R&D expenditures, treatment effects become progressively more negative moving from the bottom to the top of the spillover distribution. Patents in the lowest spillover quartile exhibit little evidence of a decline, whereas the largest and most statistically significant reductions occur among patents belonging to the highest spillover quartile. The middle quartiles display substantially smaller and statistically insignificant effects.

This pattern indicates that the average decline in spillovers documented earlier is not driven by a broad-based reduction affecting all inventions equally. Instead, stronger contractor ownership disproportionately reduced the production of inventions generating the greatest knowledge diffusion across technological fields, precisely those with the largest potential social impact.

An equally informative result emerges once flexible controls for firms' pre-treatment R&D expenditures are introduced. In the preferred specification, treatment effects become nearly identical across spillover quartiles. This occurs because allowing treatment effects to vary flexibly across firms with different research intensities absorbs the cross-quartile differences observed in the earlier specifications. The distributional pattern therefore appears to reflect differences across firms rather than differences across inventions alone, suggesting that the reallocation of innovative effort is closely linked to firms' underlying research capabilities. The next subsection examines this channel directly.

9.2 Which Firms Drive the Reallocation?

The preceding analysis established that the decline in spillovers is concentrated among inventions that historically generated the greatest external technological influence. An important remaining question is which firms account for this change. If stronger contractor ownership altered firms' research incentives, the response need not be uniform across innovators with different research capabilities.

To investigate this question, Table 10 reports treatment effects on patent production separately by spillover quartile and firms' pre-treatment R&D expenditures. Firms are partitioned into quartiles according to their average R&D spending before the reform.

The response differs sharply across firms. Among firms in the lowest R&D quartile, contractor ownership is associated with a pronounced decline in the production of patents belonging to the highest spillover quartile. By contrast, firms with the largest research programs substantially

Table 10: Patent Production by Spillover Quartile and Pre-Treatment R&D Intensity

	Pre-treatment R&D quartile				Heterogeneity
Spillover Quartile	Q1 (Low)	Q2	Q3	Q4 (High)	p-value
Top 25%	-0.814***	-0.035	0.128	0.358	0.038
50–75%	-0.018	-0.014	0.526	0.785*	0.270
25–50%	0.341	-0.065	0.271	1.895***	0.000
0–25%	0.289	-0.099	0.653**	1.298**	0.031

Notes: The table reports treatment effects from Poisson difference-in-differences specifications estimated separately for patents belonging to different quartiles of the cross-technology citation distribution. Firms are partitioned according to pre-treatment R&D expenditures. The reported coefficients correspond to the preferred specification including firm–technology fixed effects, CPC–year fixed effects, SIC2–year fixed effects, and flexible controls for pre-treatment characteristics. Standard errors are clustered at the firm level. The heterogeneity test reports the p-value from a test of equality across the four R&D-bin coefficients within each spillover quartile. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

expand patent production, with the strongest increases occurring in the lower and middle portions of the spillover distribution. Joint tests reject equality of treatment effects across R&D groups for all but one spillover quartile, indicating that these differences are statistically meaningful.

These results suggest that the decline in spillovers documented earlier does not reflect a uniform contraction in inventive activity. Instead, contractor ownership induced a reallocation of innovative effort across firms. Resource-constrained firms reduced production of the inventions generating the broadest technological influence, while firms with greater research capacity expanded patenting at the low-half of the spillover distribution. The evidence therefore points to a compositional shift in innovation rather than a decline in innovation itself.

9.3 Private Returns to Innovation

The preceding analyses indicate that contractor ownership changed the composition of innovation. A natural implication of the conceptual framework is that such a reallocation should be accompanied by an increase in the private returns to innovation. If stronger ownership rights make commercially appropriable research relatively more attractive, firms should shift innovative effort toward inventions generating greater private value.

To examine this prediction, Table 11 reports treatment effects on the patent market-value measure of Kogan et al. (2017), estimated separately by firms' pre-treatment R&D intensity.

Increases in patent market value are concentrated among firms with the largest pre-treatment

Table 11: Patent Market Value by Pre-Treatment R&D Intensity

	Patent Market Value					
	(1)	(2)	(3)	(4)	(5)	(6)
Treated×Post (<\$7.5M)	0.1512	0.1370	0.1001	0.1674	0.1192	0.2362
	(0.3641)	(0.3446)	(0.3610)	(0.3261)	(0.3342)	(0.3615)
Treated×Post (\$7.5-64M)	-0.0316	-0.0230	-0.0012	0.0339	0.0509	0.0805
	(0.1227)	(0.1231)	(0.1248)	(0.1336)	(0.1296)	(0.1380)
Treated×Post (\$64-240M)	0.0357	0.2807	0.2575	0.2718	0.2817	0.2726
	(0.3979)	(0.3598)	(0.3518)	(0.3702)	(0.3656)	(0.3471)
Treated×Post (>\$240M)	0.8358***	0.8174***	0.7811***	0.7586***	0.7359***	0.8611***
	(0.1537)	(0.1630)	(0.1757)	(0.1480)	(0.1492)	(0.1121)
Full FE	✓	✓	✓	✓	✓	✓
Patent Creativity		✓	✓	✓	✓	✓
Market Capitalization			✓	✓	✓	✓
Employees				✓	✓	✓
Citations, within cpc					✓	✓
Stochastic Volatility						✓
Joint Heterogeneity	0.000	0.001	0.004	0.004	0.007	0.000
N	15,729	14,760	14,651	14,645	14,408	14,306

Notes: The dependent variable is the patent market value measure developed by Kogan et al. (2017). Full FE refers to firm–technology, technology–year, and two-digit SIC industry–year fixed effects. Standard errors are clustered at the firm level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

research programs. While treatment effects are modest and statistically indistinguishable from zero for the first three R&D quartiles, the effect becomes both economically large and statistically significant for firms in the highest R&D quartile.

Taken together, the previous two tables point to a common mechanism. Contractor ownership shifted innovative activity away from the production of inventions generating the broadest technological spillovers and toward inventions with greater private value. The firms expanding patent production following the reform are precisely those experiencing the largest gains in patent market value, suggesting that stronger ownership rights changed the composition of innovative effort rather than simply increasing or decreasing innovative activity.

The remaining analyses examine whether this reallocation is also reflected in alternative measures of technological influence and inventive novelty that are independent of citation patterns.

9.4 Technological Significance

To examine whether the mechanism traced previously appears using a text-based measure, I next consider the patent significance measure constructed from the patent-pair data of Kelly et al. (2021). As in the main results, I exclude patent pairs assigned to the same firm in order to focus on external technological influence. I also use the count of highly similar patents rather than similarity-weighted measures, as the count-based measure more directly captures the breadth of external technological influence and therefore aligns more closely with the research question.

Table 12: Technological Significance Across the Distribution

		Patent Significance (counts, excl. self-pairs)				
A: 0-25%	Treated x Post	0.274	0.408**	0.458*	0.511**	0.590
		(0.185)	(0.208)	(0.252)	(0.244)	(.)
	N	7788	7112	7112	7028	6946
B: 25-50%	Treated x Post	-0.091	-0.073	-0.254*	-0.342**	-0.394***
		(0.110)	(0.125)	(0.145)	(0.152)	(0.143)
	N	17397	15853	15836	15625	15163
C: 50-75%	Treated x Post	-0.312***	-0.360***	-0.225*	-0.177	-0.131
		(0.117)	(0.128)	(0.117)	(0.116)	(0.119)
	N	17986	16405	16367	16195	15816
D: 75-100%	Treated x Post	-0.228*	-0.333***	-0.218	-0.214	-0.114
		(0.116)	(0.127)	(0.145)	(0.138)	(0.142)
	N	17292	15690	15672	15556	15175
Full FE		✓	✓	✓	✓	✓
Patent Creativity			✓	✓	✓	✓
Employees				✓	✓	✓
Citations, within cpc					✓	✓
R&D Expenditure						✓

Notes: The dependent variable is the reconstructed patent significance measure, defined as the count of high-similarity patent pairs excluding pairs assigned to the same firm. Full FE refers to firm–technology, technology–year, and two-digit SIC industry–year fixed effects. Standard errors are clustered at the firm level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 12 reports treatment effects separately across the distribution of this measure. The distributional decomposition broadly reinforces the evidence from cross-technology citations. Negative treatment effects extend across the upper three quartiles of the significance distribution, while the lowest quartile exhibits positive coefficients. This pattern should be interpreted in light of the underlying sample construction. Because the patent-pair data include only observations with textual similarity exceeding 50 percent, all four quartiles correspond to an exceptionally selective subset of inventions. Consequently, even the lowest quartile represents patents generating unusually high technological similarity rather than ordinary follow-on innovation. Viewed from this perspective, the results remain consistent with the citation-based evidence. Rather than indicating a broad decline in technological significance, they suggest that contractor ownership reduced the production of inventions generating highly influential follow-on research throughout the upper tail of the technological significance distribution.

The positive estimates for the lowest quartile should be interpreted cautiously. They are estimated imprecisely, and the preferred specification does not report a standard error because the variance-covariance matrix is singular, suggesting limited identifying variation within this narrowly defined subset of patents.

Table 13 next examines whether these effects differ across firms' pre-treatment R&D intensity. The cross-firm evidence closely parallels the results obtained for spillovers. Firms in the lower R&D quartiles experience economically large declines in the number of subsequent highly similar inventions generated by their patents, whereas effects among firms with the largest research programs are small and statistically indistinguishable from zero. Thus, while high-R&D firms expand patent production following the reform, there is little evidence that they simultaneously increase the external technological significance of their inventions.

9.5 Inventive Novelty

The previous analyses examined the subsequent technological influence of inventions through forward citations and forward text similarity. I conclude by considering a complementary measure that characterizes the invention itself. Table 14 reports treatment effects on the patent creativity index developed by Kalyani (2024), which is constructed solely from the textual content of the focal patent. Unlike the previous measures, the creativity index does not rely on subsequent innovation and therefore provides complementary evidence on how contractor ownership changed the nature of inventions at the time they were produced.

Table 13: High-Similarity Patent-Pair Counts by Pre-Treatment R&D Intensity

	Patent Significance (counts, excl. self-pairs)			
	(1)	(2)	(3)	(4)
Treated×Post (<$7.5M)	-0.3305**	-0.2772**	-0.2749**	-0.2542*
	(0.1492)	(0.1373)	(0.1381)	(0.1448)
Treated×Post ($7.5-64M)	-0.1737**	-0.1775**	-0.1883**	-0.1787**
	(0.0680)	(0.0690)	(0.0761)	(0.0768)
Treated×Post ($64-240M)	0.1415*	0.1175	0.1173	0.1011
	(0.0848)	(0.1022)	(0.1163)	(0.1078)
Treated×Post (>$240M)	-0.1396	0.0433	0.0928	0.0775
	(0.1263)	(0.0950)	(0.0967)	(0.0955)
Full FE	✓	✓	✓	✓
Patent Creativity		✓	✓	✓
Employees			✓	✓
Citations, within cpc				✓
Joint Heterogeneity	0.008	0.014	0.017	0.039
N	19230	17806	17806	17538

Notes: Columns report estimates separately by pre-treatment R&D quartile. Full FE refers to firm–technology, technology–year, and two-digit SIC industry–year fixed effects. Standard errors are clustered at the firm level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Average treatment effects on patent creativity are small and statistically insignificant. Once heterogeneity by firms' research intensity is considered, however, a familiar pattern emerges. Creativity declines among firms in the lowest R&D quartile but increases among firms with larger research programs. Although the creativity index captures a different dimension of innovation than either citations or patent similarity, its heterogeneity closely parallels the previous findings. The evidence therefore suggests that the reform altered the allocation of innovative effort more broadly rather than merely changing citation behavior or patterns of technological diffusion.

These findings also help interpret the creativity index within the context of this paper. Because the measure is constructed from patent text, increases in measured creativity need not reflect frontier technological advances alone but may also capture changes in the codification of technological knowledge. The positive effects among high-R&D firms are therefore difficult to reconcile with a

Table 14: Patent Creativity by Pre-Treatment R&D Intensity

	Patent Creativity				
	(1)	(2)	(3)	(4)	(5)
Treated×Post (Bin 1)	-0.178	-0.222*	-0.321***	-0.321***	-0.234*
	(0.144)	(0.127)	(0.123)	(0.123)	(0.129)
Treated×Post (Bin 2)	-0.010	-0.034	-0.064	-0.064	-0.032
	(0.069)	(0.064)	(0.063)	(0.063)	(0.071)
Treated×Post (Bin 3)	0.037	0.029	-0.027	-0.027	0.038
	(0.084)	(0.088)	(0.109)	(0.109)	(0.111)
Treated×Post (Bin 4)	0.333***	0.367***	0.310***	0.310***	0.360***
	(0.095)	(0.099)	(0.114)	(0.114)	(0.124)
Industry-Year FE	✓	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓	✓
Employees		✓	✓	✓	✓
Patents filed			✓	✓	✓
Agegroup				✓	✓
# of CPCs					✓
Joint Heterogeneity	0.009	0.001	0.002	0.002	0.007
SE cluster	firm	firm	firm	firm	firm
N	17710	17707	17707	17707	17086

Notes: The dependent variable is patent creativity, measured using the index developed by Kalyani (2024). Columns report estimates separately by pre-treatment R&D quartile. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

broad increase in breakthrough innovation, given that these firms primarily expand production at the bottom half of the spillover distribution. Instead, they are consistent with the broader evidence that stronger contractor ownership redirected innovative effort toward research with greater private appropriability and more codified technological content.

The evidence presented in this section consistently points toward a change in the composition of innovation rather than a change in the diffusion process itself. The 1983 reform did not alter the public disclosure requirements of the patent system; patented inventions continued to become publicly available upon grant. Instead, its central innovation was to make contractor ownership the prospective default allocation of patent rights for federally funded inventions. By allowing firms to anticipate ownership before undertaking research, the reform changed the relative profitability of alternative research projects, strengthening incentives to pursue inventions whose returns could be more fully appropriated.

The empirical patterns align closely with this mechanism. The reform disproportionately reduced the production of inventions generating the broadest technological spillovers, particularly among firms with limited research resources. At the same time, firms with the largest research programs expanded patenting to the bottom half of the spillover distribution and experienced the largest increases in patent market value. Alternative text-based measures of technological significance and novelty reinforce the same conclusion.

Taken together, the evidence suggests that expanding contractor ownership influenced not only who captured the returns from publicly funded research but also the direction of innovative effort itself. Rather than making knowledge more excludable after inventions were produced, stronger ownership rights changed which inventions firms chose to pursue in the first place.

10 Robustness Tests

This section examines whether the estimated decline in knowledge spillovers is sensitive to the construction of the comparison group or to the estimation strategy. The first two exercises modify the set of eligible control observations before matching, while the final exercise replaces the baseline difference-in-differences estimator with the Synthetic Difference-in-Differences (SDID) estimator of Arkhangelsky et al. (2021).

Table 15 restricts the donor pool prior to matching to control firm–technology cells whose pre-treatment characteristics lie within the central 98 percent of the treated distribution. Specifically, I exclude potential controls outside the 1st–99th percentiles of the treated distribution in firm size, technological breadth, patenting activity, R&D expenditures, patent creativity, and pre-treatment spillover levels and trends. Across specifications, the estimated effect of contractor ownership remains negative and close in magnitude to the baseline estimates, indicating that the results are not driven by a small number of economically distant control observations.

Table 15: Spillovers Excluding Tail Control Matches

	Spillovers						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated x Post	-0.2445***	-0.2304***	-0.1913**	-0.1977**	-0.2445***	-0.1868**	-0.2075**
	(0.0749)	(0.0712)	(0.0787)	(0.0803)	(0.0826)	(0.0886)	(0.0902)
Industry-Year FE	✓	✓	✓	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓	✓	✓	✓
Patent Creativity		✓	✓	✓	✓	✓	✓
Employees			✓	✓	✓	✓	✓
Patents filed				✓	✓	✓	✓
R&D					✓	✓	✓
# of CPCs						✓	✓
Agegroup							✓
SE	firm	firm	firm	firm	firm	firm	firm
N	20,835	15,579	15,570	15,570	15,545	15,539	15,539

Notes: The donor pool is restricted prior to matching to control firm–technology cells lying within the 1st–99th percentiles of the treated distribution for pre-treatment firm size (employees), technological breadth (number of CPC classes), patenting activity, R&D expenditures, patent creativity, spillovers, and spillover trends. Matching and estimation otherwise follow the baseline specification. Standard errors are clustered at the firm level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Large federal contractors differ systematically from the average patenting firm. To ensure that the baseline estimates are not driven by comparisons with substantially smaller firms, Table 16 restricts the donor pool to control firm–technology cells whose pre-treatment employment exceeds the 25th percentile of treated firms. The estimated decline in spillovers remains statistically significant across specifications and, if anything, is somewhat larger than in the baseline estimates.

Table 16: Spillovers Using Comparable-Sized Control Firms

	Spillovers						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated x Post	0.1073	-0.1945	-0.2464*	-0.2619*	-0.2888**	-0.4025**	-0.4111**
	(0.1175)	(0.1209)	(0.1324)	(0.1474)	(0.1283)	(0.1922)	(0.1918)
Industry-Year FE	✓	✓	✓	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓	✓	✓	✓
Patent Creativity		✓	✓	✓	✓	✓	✓
Employees			✓	✓	✓	✓	✓
Patents filed				✓	✓	✓	✓
R&D					✓	✓	✓
# of CPCs						✓	✓
Agegroup							✓
SE	firm	firm	firm	firm	firm	firm	firm
N	21,968	13,513	13,113	13,113	12,969	12,802	12,802

Notes: The donor pool is restricted prior to matching to control firm–technology cells whose pre-treatment employment is at least as large as the 25th percentile of treated firms. Matching and estimation otherwise follow the baseline specification. Standard errors are clustered at the firm level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

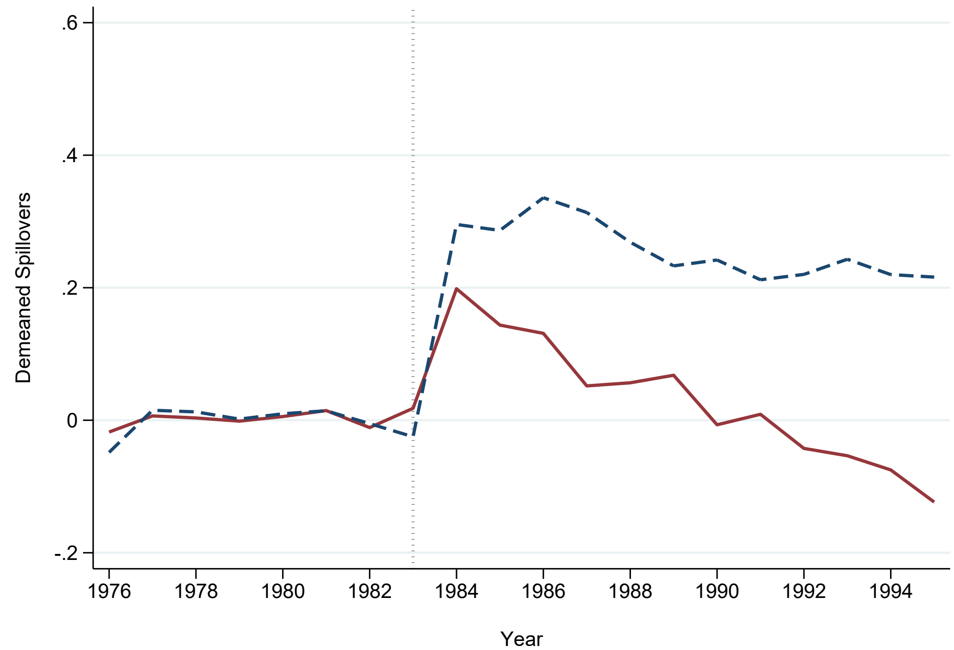
The baseline specification combines matching with a difference-in-differences estimator. As an alternative approach, Table 17 re-estimates the treatment effect using the Synthetic Difference-in-Differences estimator of Arkhangelsky et al. (2021), which constructs a weighted control group to reproduce the pre-treatment evolution of treated observations. The estimated treatment effects remain negative and statistically significant under both the matched donor pool and the full-sample specification, indicating that the results are not driven by the particular weighting scheme implied by the baseline estimator. Donor weight diagnostics indicate that donor weights are broadly distributed across the donor pool, with no single observation receiving disproportionate weight (available in Appendix table A14).

Table 17: Synthetic Difference-in-Differences: Spillovers

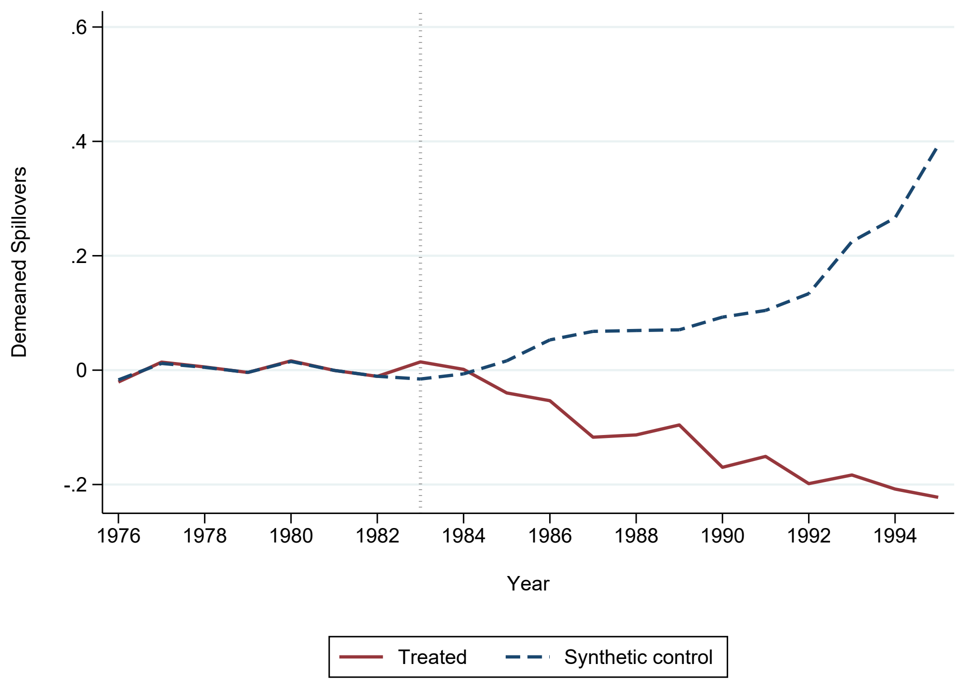
	Spillovers	
	Matched donor pool	Full sample
Treatment effect	-0.216^{***}	-0.230^{***}
	(0.0233)	(0.0217)
Treated units	1,598	1,736
Donor units	3,003	14,008
Observations	92,020	314,880

Notes: The table reports estimates from the Synthetic Difference-in-Differences (SDID) estimator of Arkhangelsky et al. (2021). Column (1) uses the matched donor pool from the baseline analysis after deduplication of matched controls. Column (2) uses the full sample of eligible donor units. Inference is based on 500 placebo replications. Standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 8 plots the corresponding treatment paths. In both panels, treated and synthetic control units closely track one another throughout the pre-treatment period. Following the 1983 reform, the two series diverge persistently, with treated firm–technology cells exhibiting substantially lower spillovers than their synthetic counterparts. Taken together, these exercises indicate that the estimated decline in spillovers is not sensitive to the construction of the comparison group or to the estimation strategy. Across all specifications, the estimated effects remain economically large and statistically significant.



(a) Matched donor pool



(b) Full sample

Figure 8: Synthetic Difference-in-Differences Paths

Notes: The figure plots the treated and synthetic control paths estimated by the Synthetic Difference-in-Differences estimator of Arkhangelsky et al. (2021). The outcome is the demeaned log of spillovers, where demeaning subtracts the average outcome over the 1976–1982 pre-treatment period within each firm–technology cell. The vertical dashed line denotes the 1983 policy reform.

11 Conclusion

This paper studies how the prospective allocation of intellectual-property rights over publicly funded inventions shapes firms' research choices and the subsequent diffusion of technological knowledge. I exploit the 1983 Presidential Memorandum on Government Patent Policy, which substantially expanded contractors' ability to retain ownership of future federally funded inventions. At the same time, the underlying research funding mechanism and the public disclosure requirements of the patent system remained largely unchanged. By prospectively increasing the expected private returns to successful innovation, the reform provides a natural experiment for studying how ownership rights affect the direction and social reach of innovative activity.

The evidence shows that expanding contractor ownership reduced the external technological influence of patented inventions. Following the reform, treated firm–technology units generated innovation associated with fewer cross-technology spillovers, lower rates of technologically similar follow-on innovation outside the originating firm, and higher private market value. These findings are robust across alternative measures of knowledge diffusion, alternative definitions of external spillovers, and a synthetic difference-in-differences estimator. Because the reform left the public disclosure regime governing patented inventions unchanged, the evidence is more naturally interpreted as reflecting changes in the composition of innovation than changes in the diffusion of a given invention.

The paper also provides evidence on the mechanism underlying these changes. Rather than uniformly reducing innovative activity, stronger ownership rights altered the composition of research. Firms became relatively less likely to produce inventions in the upper tail of the spillover distribution and more likely to generate innovations with narrower external applicability. These compositional changes are concentrated among firms with high research intensity, consistent with appropriability changing the relative profitability of research projects rather than aggregate inventive effort.

These findings provide causal evidence on the fundamental tradeoff between stronger private appropriability and the diffusion of technological knowledge. The partially public-good nature of technological knowledge implies that stronger appropriability can simultaneously increase private incentives to innovate while reducing the knowledge available to future innovators. The empirical evidence supports this prediction: expanding contractor ownership increased the private returns to innovation but reduced their external technological influence. The evidence therefore suggests that stronger appropriability narrows Arrow's private–social wedge from both directions: by increasing the private returns to innovation while simultaneously reducing the external benefits generated by new knowledge.

From a policy perspective, the paper highlights ownership allocation as an important but often underappreciated dimension of innovation policy. Policies governing who owns publicly funded inventions do more than determine the distribution of returns after discoveries are made; they shape the kinds of discoveries firms choose to pursue in the first place. By changing the relative profitability of research projects with different private and social returns, ownership rules influence the balance between private appropriation and knowledge diffusion. The allocation of intellectual-property rights over publicly funded research should therefore be viewed not merely as a legal or administrative choice, but as a policy instrument that helps determine the social returns generated by public R&D investment.

Finally, the analysis points to important dynamic implications. By shifting innovative effort toward technologies with narrower applicability, stronger ownership rights alter the composition of the knowledge accumulated by firms, leading to a more specialized knowledge base. As firms increasingly build capabilities in narrower technological domains, the stock of broadly applicable knowledge grows more specialized, limiting the range of ideas that future innovators can combine and extend. At the same time, generating fewer spillovers leaves firms with fewer opportunities to benefit from knowledge created elsewhere. To the extent that the returns to R&D are affected by the availability of external knowledge, as emphasized by the literature on absorptive capacity, these dynamics may also weaken incentives for future R&D investment and, over time, the development of firms' absorptive capacity. More broadly, these dynamic effects suggest that ownership policies governing publicly funded research may influence not only the contemporaneous allocation of private and social returns from innovation, but also the long-run accumulation and composition of the knowledge base that underlies technological progress and productivity growth.

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Appendix

A Data Construction

A.1 Patent ownership data

The ownership and government funding information used throughout the paper comes from the patent-level dataset assembled by Gross and Sampat (2025). Their contribution substantially expands earlier compilations of federally funded patents by exploiting a previously unused administrative source preserved in the USPTO archives. These archival records identify whether title to a federally funded invention remained with the federal government or was assigned to the contractor while the government retained a non-exclusive license. For a subset of patents, the records also identify the funding agency.

The ownership distinction is the central institutional variable in this paper. Government-title patents correspond to inventions for which ownership remained with the sponsoring federal agency, whereas contractor-title ("license") patents correspond to inventions for which ownership was assigned to the performing organization subject to the government's retained rights. Throughout the paper, I adopt the ownership classifications provided in Gross and Sampat (2025), except for the treatment of post-1983 indeterminate observations discussed in Section A.4.

Although the full ownership dataset spans patents filed between 1950 and 2019, the empirical analysis exploits variation generated by the 1983 policy reform and therefore restricts estimation to patents filed between 1976 and 1995. Earlier and later years are nevertheless retained whenever necessary to construct citation- and text-based outcome variables.

A.2 Patent–firm matching

Patent ownership information is linked to publicly listed firms using the patent-to-Compustat linkage developed by Dyevre (2023). Their algorithm combines information from PatentsView, historical assignee names, Compustat, and corporate ownership records to assign patents to firms while accounting for mergers, acquisitions, name changes, and other corporate reorganizations.

For the purposes of this paper, I use the version of the linkage that assigns patents to the firm's ultimate owner at the time of filing rather than to subsequent owners following corporate

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restructuring. This choice aligns patent ownership with the firm that undertook the underlying research investment and therefore faced the incentives generated by changes in innovation policy.

Because identifying federally funded inventions is particularly important for the empirical analysis, I extend the Dyevre (2023) linkage by matching previously unmatched patents to Compustat firms. Candidate matches are generated using the disambiguated assignee names available in PatentsView together with large-language-model assistance and are subsequently verified manually before inclusion in the final dataset.

This extension adds 70,679 patent-to-firm matches over the full 1950–2019 sample, including 7,147 federally funded patents. Restricting attention to the 1976–1995 estimation period, the extension contributes 17,484 additional patent matches, including 2,931 federally funded patents. While these observations account for only about 7% of patents in the final estimation sample, they increase the number of matched federally funded patents by more than 50% relative to the original linkage.

A.3 Assignee classification

The ownership records in Gross and Sampat (2025) identify the legal ownership status of federally funded patents but do not classify assignees by organizational type. Because the institutional reforms studied in this paper affected different organizations at different times, I construct additional assignee classifications using the disambiguated assignee names available in PatentsView.

Each assignee is classified into one of four broad organizational categories: business firms, universities, government agencies, and individuals. In addition, I separately identify patents jointly assigned to firms and universities and patents jointly assigned to firms and government agencies. These mixed organizational forms are useful for assessing whether the estimated effects are influenced by inventions whose research direction may have been jointly determined by organizations operating under different institutional environments.

To distinguish nonprofit organizations from business firms, I additionally classify organizations according to their tax status under Sections 501(c)(3) and 501(c)(4) of the Internal Revenue Code. Candidate classifications are generated using large-language-model assistance applied to the PatentsView assignee names and are manually verified before incorporation into the dataset.

These classifications are used primarily for robustness exercises. The baseline analysis assigns patents according to filing-date corporate ownership, while alternative specifications progressively exclude patents involving universities, nonprofit organizations, government agencies, and mixed organizational forms.

A.4 Treatment of indeterminate ownership observations

A small fraction of federally funded patents in Gross and Sampat (2025) are classified as indeterminate because the available archival records do not permit ownership status to be assigned unambiguously. These cases arise when the patent documents do not clearly identify whether title remained with the government or was assigned to the contractor, or when the archival material contains conflicting or incomplete ownership information.

For patents filed before the 1983 policy reform, I follow the ownership classifications in Gross and Sampat (2025) and exclude indeterminate observations from the estimation sample. These patents constitute only a very small share of pre-reform federally funded patents, and their exclusion has no meaningful effect on the composition of the sample.

The post-1983 period requires different treatment. Following correspondence with the authors of Gross and Sampat (2025), indeterminate patents assigned to business firms should, for practical purposes, be interpreted as contractor-owned under the post-reform institutional regime. The public release retains these observations as indeterminate in order to remain faithful to the original archival records rather than impose an additional interpretation.

Accordingly, the baseline analysis classifies post-1983 indeterminate patents assigned to firms as contractor-owned ("license" in the Gross–Sampat coding). This reassignment affects 1,149 federally funded patents in the estimation sample. This treatment is both institutionally and empirically motivated. Institutionally, it follows the interpretation recommended by the authors of Gross and Sampat (2025), who note that these observations remain classified as indeterminate in the public release solely to preserve fidelity to the archival records rather than because they correspond to a distinct ownership regime. Empirically, inspection of the disambiguated assignee names reveals that only 19 of these patents involve non-firm co-assignees, all of which are joint assignments between firms and universities. Excluding these 19 observations leaves the estimates virtually unchanged, indicating that the baseline results are not driven by mixed-assignee cases.

A.5 Citation construction

Patent citations are assembled from two complementary sources. For patents filed before 1976, I use the historical citation data constructed by **?**, which recover citation links for the early U.S. patent system. For patents filed from 1976 onward, citation information is obtained directly from the USPTO through PatentsView.

Combining the two datasets yields a harmonized citation panel spanning patents filed between 1950 and 2019. The extended time horizon is important because forward citations accumulate gradually over time and because patents filed near the beginning or end of the estimation window require observations outside 1976–1995 to measure subsequent knowledge flows consistently.

The baseline spillover measure counts forward citations originating from patents assigned to different CPC technology classes than the cited patent. Unless otherwise noted, citations are accumulated over a ten-year forward window following the filing year. Alternative citation windows and alternative spillover definitions are examined as robustness exercises.

A.6 Innovation outcome measures

The paper employs four complementary classes of innovation outcomes that capture different dimensions of research performance.

Citation-based measures. The primary outcomes measure knowledge diffusion using forward patent citations. At the patent level, spillovers are measured as forward citations originating from patents outside the focal patent's own CPC technology class. At the firm–technology–year level, these citations are aggregated across all patents filed within the same firm and technology cell. Additional outcomes exclude self-citations to isolate external knowledge diffusion.

Text-based measures. To complement citation-based outcomes, I use the patent-level textual similarity measures developed by Kelly et al. (2021). Their publicly available dataset provides measures of patent impact and novelty derived from similarities between patent texts. I additionally use the authors' patent-pair similarity data for patent pairs with textual similarity exceeding 50%, which permit construction of measures distinguishing within-firm knowledge reuse from diffusion to external innovators. These pair-level measures form the basis of the external diffusion outcomes reported in the paper.

Patent creativity. I also employ the patent creativity index developed by Kalyani (2024), which measures novelty through new combinations of text bi-grams appearing in patent documents. Relative to citation-based measures and the full-text similarity measures of Kelly et al. (2021), this index captures a different dimension of inventive activity and is available for nearly the entire estimation sample. In the context of this paper, the measure is best interpreted as reflecting a combination of technological novelty and the codification of knowledge into patentable form. Because both processes were already evolving prior to the 1983 reform, the empirical specifications control flexibly for pre-treatment differences in creativity across firm–technology cells. The creativity measure is therefore not interpreted as a standalone estimate of treatment effects but rather as an independent source of variation for evaluating whether the broader pattern of results is consistent across alternative measures of innovation.

Market value. Finally, I measure the private value of innovation using the patent valuation estimates of Kogan et al. (2017). These estimates infer the value of individual patents from abnormal stock-market reactions around patent grants. Although coverage is limited to publicly traded firms with available financial-market data, the market-value measure provides a useful complement to citation- and text-based outcomes by directly capturing the private economic returns to patented inventions.

A.7 Construction of the estimation sample

The empirical analysis is conducted at the firm–technology–year level, where technologies are defined using three-digit Cooperative Patent Classification (CPC) classes. Patents are assigned to firms according to filing-date ownership and then aggregated within firm, technology, and filing year.

Treatment status is determined entirely using the pre-policy period. A firm–technology cell is classified as treated if it receives at least one federally funded patent before the 1983 policy reform. Control observations consist of firm–technology cells that never receive federally funded patents during the sample period. Firm–technology cells first receiving federal funding after 1983 are excluded from the estimation sample in order to maintain a clean comparison between pre-existing recipients and never-treated firms.

The baseline estimation window spans 1976–1995. This period provides sufficiently long pre- and post-reform horizons while avoiding major institutional changes to the U.S. patent system during the late 1990s that could otherwise confound the interpretation of the policy effect.

B Figures

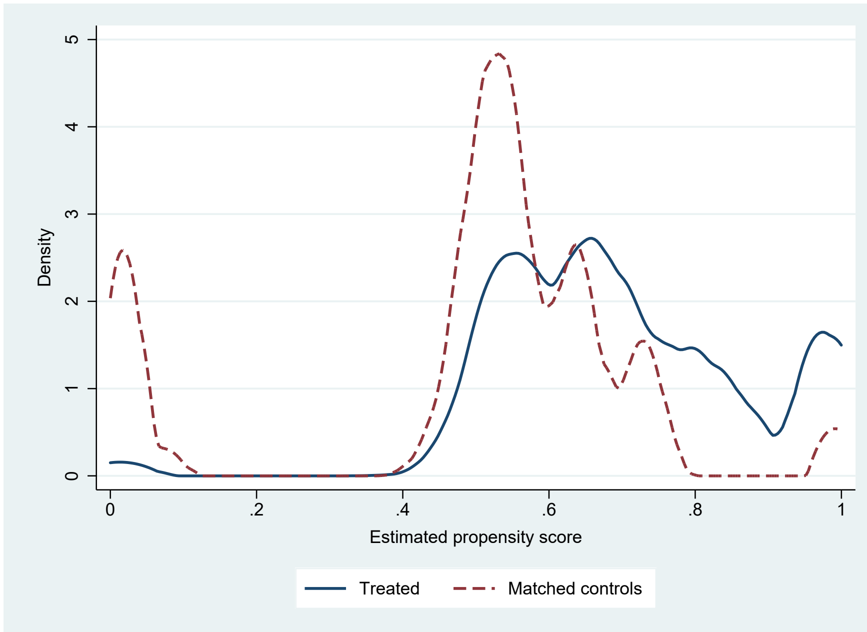

Figure A1: Common Support of Propensity Scores of Matched Groups

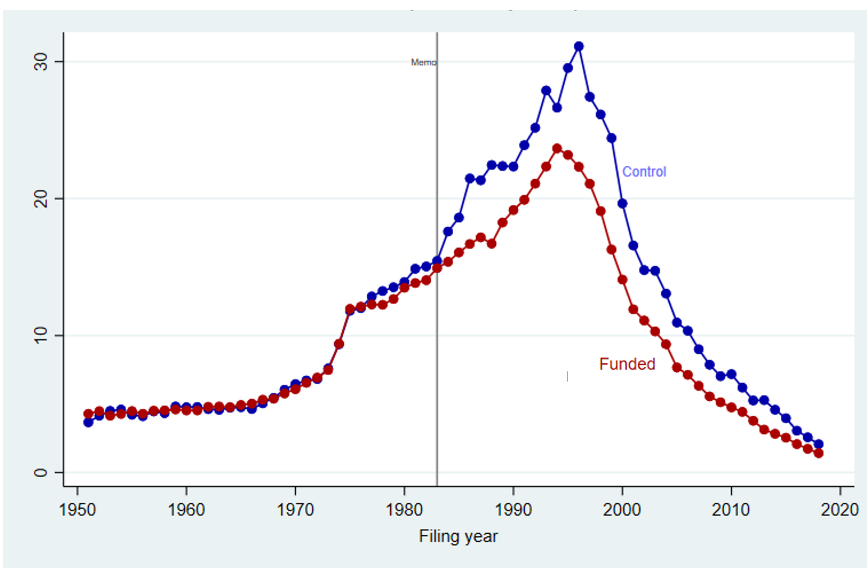

Figure A2: Spillovers by Matched Group, 1950–2019

Notes: The figure plots average cross-technology forward citations for treated and matched control firm-technology cells by filing year. The vertical line marks the 1983 reform.

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C Tables

C.1 Descriptive Evidence

Table A1: Spillovers & Patent Ownership , 1950-2019, (excl. self-citations)

	Spillovers						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Govt-funded	0.302***	0.268***	0.269***	0.250***			
	(0.032)	(0.033)	(0.030)	(0.031)			
License	-0.227***	-0.183***	-0.166***	-0.128***	-0.162***	-0.132**	-0.095*
	(0.038)	(0.038)	(0.035)	(0.037)	(0.046)	(0.052)	(0.055)
N	2,002,341	2,002,317	2,001,677	2,001,653	25,426	24,415	22,730
Patent Controls	Y	Y	Y	Y	Y	Y	Y
Firm Controls	Y	Y	Y	Y	Y	Y	Y
Year FE	✓	✓			✓		
CPC FE	✓	✓			✓		
Industry FE		✓		✓			
CPC-Year FE			✓	✓		✓	
Agency FE					✓	✓	
Agency-CPC-Year FE							✓
SE	cpc-year	cpc-year	cpc-year	cpc-year	cpc-year	cpc-year	cpc-year

Notes: Spillovers defined as cross-CPC citations over next ten years excluding self-citations and assumed to be Poisson distributed. Controls: citations over next 10 years, TFP as sales/employees, R&D productivity as R&D expenditure over patents-per-year. Standard errors clustered at the year-cpc_class level. Year FE refer to the filing year. (* p <0.10, ** p <0.05, *** p <0.01)

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C.2 Main Results

Table A2: Effects on Cross-Technology Spillovers (firms only)

	Spillovers						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated x Post	-0.071	-0.265***	-0.230***	-0.215**	-0.247***	-0.317***	-0.314***
	(0.110)	(0.086)	(0.088)	(0.090)	(0.087)	(0.108)	(0.108)
Industry-Year FE	✓	✓	✓	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓	✓	✓	✓
Patent Creativity		✓	✓	✓	✓	✓	✓
Employees			✓	✓	✓	✓	✓
Patents filed				✓	✓	✓	✓
R&D					✓	✓	✓
# of CPCs						✓	✓
Agegroup							✓
SE	firm	firm	firm	firm	firm	firm	firm
N	24,783	19,198	19,181	19,181	18,448	17,838	17,838

Notes: Spillovers are defined as cross-CPC forward citations over the next ten years and assumed to be Poisson distributed. Patents with universities, state agencies or NGOs as co-assignees are excluded. All controls refer to pre-treatment quartile bins allowing for flexible time paths post-1983. Patent creativity is a measure of novelty of a patent developed by Kalyani (2024). (* p <0.10, ** p <0.05, *** p <0.01)

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Table A3: Effects on Cross-Technology Spillovers (excl. self-citations)

	Spillovers						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated x Post	-0.078	-0.182**	-0.227**	-0.207**	-0.271***	-0.345***	-0.366***
	(0.084)	(0.077)	(0.091)	(0.096)	(0.089)	(0.108)	(0.107)
Industry-Year FE	✓	✓	✓	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓	✓	✓	✓
Pat Creat'ty		✓	✓	✓	✓	✓	✓
Employees			✓	✓	✓	✓	✓
Patents filed				✓	✓	✓	✓
R&D					✓	✓	✓
# of CPCs						✓	✓
Agegroup							✓
SE	firm	firm	firm	firm	firm	firm	firm
N	26,224	20,032	19,991	19,991	19,252	18,817	18,817

Notes: Spillovers are defined as cross-CPC forward citations over the next ten years and assumed to be Poisson distributed. Identified same-firm citations are excluded. All controls refer to pre-treatment quartile bins allowing for flexible time paths post-1983. Patent creativity is a measure of novelty of a patent developed by Kalyani (2024). (* p <0.10, ** p <0.05, *** p <0.01)

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Table A4: Effects on Cross-Technology Spillovers (incl. trend-controls)

	Spillovers			
	(1)	(2)	(3)	(4)
Treated x Post	-0.328***	-0.258***	-0.258***	-0.258***
	(0.108)	(0.091)	(0.090)	(0.091)
Industry-Year FE	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓
Patent Creativity	✓	✓	✓	✓
Employees	✓	✓	✓	✓
Patents filed	✓	✓	✓	✓
R&D	✓			
# of CPCs	✓		✓	
Trends:				
Patent Creativity		✓	✓	✓
Employees			✓	✓
Patents filed				✓
SE	firm	firm	firm	firm
N	17,852	19,222	19,222	19,222

Notes: Spillovers are defined as cross-CPC forward citations over the next ten years and assumed to be Poisson distributed. All controls refer to pre-treatment quartile bins allowing for flexible time paths post-1983. Pre-treatment trends are measured as the estimated linear slope of the variable over the pre-1983 period. (* p <0.10, ** p <0.05, *** p <0.01)

C.3 Text-Based Evidence on Technological Significance

Table A5: KPST (2021) Patent Significance Measure

	Patent Significance						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated x Post	0.0107	-0.0077	-0.0057	-0.0052	-0.0110	-0.0041	-0.0062
	(0.0102)	(0.0082)	(0.0092)	(0.0091)	(0.0105)	(0.0140)	(0.0138)
Industry-Year FE	✓	✓	✓	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓	✓	✓	✓
Pat Creativity		✓	✓	✓	✓	✓	✓
Employees			✓	✓	✓	✓	✓
Patents filed				✓	✓	✓	✓
R&D					✓	✓	✓
# of CPCs						✓	✓
Agegroup							✓
SE	firm	firm	firm	firm	firm	firm	firm
N	25,262	19,363	19,323	19,323	18,584	17,910	17,910

Notes: Patent significance is measured as text similarity of a patent with future patents according to Kelly et al. (2021). This table refers to the publicly available version of the measure which is calculated as the sum of similarity indexes of all future patents, both within and across firms. Thus, firms that file more patents will mechanically score higher patent significance as their patents will naturally have high text similarity. (* p <0.10, ** p <0.05, *** p <0.01)

← Back to Tech Significance

Table A6: KPST (2021) Patent Significance Measure (weighted, firms only)

	Patent Significance						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated x Post	-0.034	-0.083***	-0.036*	-0.038*	-0.058***	-0.065**	-0.068***
	(0.022)	(0.022)	(0.020)	(0.021)	(0.022)	(0.026)	(0.026)
Industry-Year FE	✓	✓	✓	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓	✓	✓	✓
Pat Creativity		✓	✓	✓	✓	✓	✓
Employees			✓	✓	✓	✓	✓
Patents filed				✓	✓	✓	✓
R&D					✓	✓	✓
# of CPCs						✓	✓
Agegroup							✓
SE	firm	firm	firm	firm	firm	firm	firm
N	24,086	18,612	18,595	18,595	17,843	17,340	17,340

Notes: Patent significance is measured as text similarity of a patent with future patents according to Kelly et al. (2021). The outcome variable is the publicly available measure weighted by the share of external forward citations to partially account for the mechanic weights firms that patent a lot receive in the standard version of the Kelly measure. Patents with universities, state agencies or NGOs as co-assignees are excluded to account for the potential effect the earlier, 1980, grant of title rights to them might have on research decisions. (* p <0.10, ** p <0.05, *** p <0.01)

Table A7: KPST (2021) Patent Significance Measure (weighted, firms only, no zeros)

	Patent Significance						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated x Post	-0.019	-0.057***	-0.042***	-0.044***	-0.057***	-0.056***	-0.058***
	(0.017)	(0.014)	(0.015)	(0.016)	(0.017)	(0.019)	(0.019)
Industry-Year FE	✓	✓	✓	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓	✓	✓	✓
Patent Creativity		✓	✓	✓	✓	✓	✓
Employees			✓	✓	✓	✓	✓
Patents filed				✓	✓	✓	✓
R&D					✓	✓	✓
# of CPCs						✓	✓
Agegroup							✓
SE	firm	firm	firm	firm	firm	firm	firm
N	23,975	18,472	18,455	18,455	17,693	17,208	17,208

Notes: This outcome variable is identical to the one of table A6 but excluding firm-technology observations that have zero patent significance after weighting by the share of external citations. A mass of zeros may affect the estimate drawing it towards zero (* p <0.10, ** p <0.05, *** p <0.01)

Table A8: Effects on External Text-Based Patent Significance (counts)

	Patent Significance						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated x Post	-0.3340***	-0.3631***	-0.2676*	-0.2606*	-0.2636*	-0.2139	-0.2170
	(0.1228)	(0.1263)	(0.1493)	(0.1401)	(0.1406)	(0.1363)	(0.1368)
Industry-Year FE	✓	✓	✓	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓	✓	✓	✓
Patent Creativity		✓	✓	✓	✓	✓	✓
Employees			✓	✓	✓	✓	✓
Patents filed				✓	✓	✓	✓
R&D					✓	✓	✓
# of CPCs						✓	✓
Agegroup							✓
SE	firm	firm	firm	firm	firm	firm	firm
N	25,327	19,270	19,227	19,227	18,492	17,847	17,847

Notes: Re-constructed KPST measure calculating forward and backward similarity by summing over counts of pairs with similarity above 50%. Identified same-owner pairs are excluded. (* p <0.10, ** p <0.05, *** p <0.01)

C.4 External Diffusion Share

Table A9: Effects on the External Diffusion Share (weights)

	Externalization Share				
	(1)	(2)	(3)	(4)	(5)
Treated x Post	-0.105**	-0.152***	-0.153***	-0.149***	-0.166***
	(0.041)	(0.045)	(0.045)	(0.044)	(0.043)
Industry-Year FE	✓	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓	✓
Pat Creat'ty		✓	✓	✓	✓
Employees			✓	✓	✓
Patents filed				✓	✓
R&D					✓
Agegroup					
SE	firm	firm	firm	firm	firm
N	24,081	18,471	18,433	18,433	17,903

Notes: External diffusion is measured as the share of text similarity of a patent with external future patents over total similarity of future patents. This table refers to the re-constructed version of the measure from the high-similarity dataset provided by the authors. Text similarity is calculated following Kelly et al. (2021) as the sum of similarity indexes of all future patents, both within and across firms. Thus, firms that file more patents will mechanically score higher patent significance as their patents will naturally have high text similarity. (* p <0.10, ** p <0.05, *** p <0.01)

← Back to Externalization Share

Table A10: Effects on the External & Cross-CPC Diffusion Share (counts)

	Externalization Share				
	(1)	(2)	(3)	(4)	(5)
Treated x Post	-0.110***	-0.154***	-0.143***	-0.133***	-0.142***
	(0.038)	(0.042)	(0.050)	(0.047)	(0.047)
Industry-Year FE	✓	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓	✓
Patent Creativity		✓	✓	✓	✓
Employees			✓	✓	✓
Patents filed				✓	✓
R&D					✓
Agegroup	firm	firm	firm	firm	firm
SE	23,980	18,199	18,153	18,153	17,520

Notes: External diffusion is measured as the share of text similarity of a patent with external future patents over total similarity of future patents. This table refers to the re-constructed version of the measure from the high-similarity dataset provided by the authors. Text similarity is calculated following Kelly et al. (2021) as the sum of similarity indexes of all future patents, both within and across firms. Thus, firms that file more patents will mechanically score higher patent significance as their patents will naturally have high text similarity. (* p <0.10, ** p <0.05, *** p <0.01)

← Back to Externalization Share

Table A11: Effects on the External & Cross-CPC Diffusion Share (weights)

	Externalization Share				
	(1)	(2)	(3)	(4)	(5)
Treated x Post	-0.122***	-0.166***	-0.148***	-0.139***	-0.144***
	(0.039)	(0.043)	(0.048)	(0.046)	(0.046)
Industry-Year FE	✓	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓	✓
Patent Creativity		✓	✓	✓	✓
Employees			✓	✓	✓
Patents filed				✓	✓
R&D					✓
Agegroup	firm	firm	firm	firm	firm
SE	23,876	18,183	18,148	18,148	17,531

Notes: External diffusion is measured as the share of text similarity of a patent with external future patents over total similarity of future patents. This table refers to the re-constructed version of the measure from the high-similarity dataset provided by the authors. Text similarity is calculated following Kelly et al. (2021) as the sum of similarity indexes of all future patents, both within and across firms. Thus, firms that file more patents will mechanically score higher patent significance as their patents will naturally have high text similarity. (* p <0.10, ** p <0.05, *** p <0.01)

← Back to Externalization Share

Table A12: Effects on the External & Cross-CPC Diffusion Share (firms only, counts)

	Externalization Share				
	(1)	(2)	(3)	(4)	(5)
Treated x Post	-0.136***	-0.198***	-0.181***	-0.168***	-0.180***
	(0.040)	(0.044)	(0.050)	(0.048)	(0.050)
Industry-Year FE	✓	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓	✓
Patent Creativity		✓	✓	✓	✓
Employees			✓	✓	✓
Patents filed				✓	✓
R&D					✓
Agegroup	firm	firm	firm	firm	firm
SE	23,884	18,185	18,172	18,172	17,548

Notes: External diffusion is measured as the share of text similarity of a patent with external future patents over total similarity of future patents. This table refers to the re-constructed version of the measure from the high-similarity dataset provided by the authors. Text similarity is calculated following Kelly et al. (2021) as the sum of similarity indexes of all future patents, both within and across firms. Thus, firms that file more patents will mechanically score higher patent significance as their patents will naturally have high text similarity. (* p <0.10, ** p <0.05, *** p <0.01)

← Back to Externalization Share

Table A13: Effects on the External & Cross-CPC Diffusion Share (firms only, weights)

	Externalization Share				
	(1)	(2)	(3)	(4)	(5)
Treated x Post	-0.141***	-0.198***	-0.191***	-0.184***	-0.194***
	(0.040)	(0.045)	(0.052)	(0.050)	(0.052)
Industry-Year FE	✓	✓	✓	✓	✓
Tech-Year FE	✓	✓	✓	✓	✓
Firm-Tech FE	✓	✓	✓	✓	✓
Patent Creativity		✓	✓	✓	✓
Employees			✓	✓	✓
Patents filed				✓	✓
R&D					✓
Agegroup	firm	firm	firm	firm	firm
SE	23,857	18,191	18,178	18,178	17,545

Notes: External diffusion is measured as the share of text similarity of a patent with external future patents over total similarity of future patents. This table refers to the re-constructed version of the measure from the high-similarity dataset provided by the authors. Text similarity is calculated following Kelly et al. (2021) as the sum of similarity indexes of all future patents, both within and across firms. Thus, firms that file more patents will mechanically score higher patent significance as their patents will naturally have high text similarity. (* p <0.10, ** p <0.05, *** p <0.01)

← Back to Externalization Share

C.5 Synthetic Difference-in-Differences

Table A14: Synthetic D-i-D: Donor Weight Diagnostics

Treated units	Donor units	Positive donors	Mean weight	SD	Min	Median (p50)	p95	p99	Max
A. Matched Donor Pool									
1,598	3,003	3,003	0.000333	0.0000207	0.0003081	0.0003342	0.0003389	0.0004175	0.0006142
B. Full Sample									
1,736	14,008	14,008	0.0000714	0.00000764	0.0000608	0.0000709	0.0000776	0.0001025	0.000246

Notes: The table summarizes donor weights assigned by the Synthetic Difference-in-Differences (SDID) estimator on the matched donor pool. Donor units correspond to unique matched control firm–technology cells after deduplication. All donor units receive positive weight. The narrow distribution of donor weights indicates that the synthetic comparison group is constructed from a broad set of donor units rather than being driven by a small number of influential controls.

Table A15: Synthetic D-i-D: Spillovers (excl. indeterminates)

	Spillovers
ATT	-0.230^{***}
	(0.0215)
Observations	314680

Notes: The table reports estimates from the Synthetic Difference-in-Differences (SDID) estimator of Arkhangelsky et al. (2021). Estimation uses the full sample of eligible donor units. Inference is based on 500 placebo replications. Standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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